Computational Fluid Dynamics (CFD)

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Introduction

The lid-driven cavity problem is one of the classical benchmark problems in fluid mechanics and is widely used to evaluate numerical methods. It is commonly employed to study the fundamental characteristics of incompressible flow in a confined cavity driven by a moving lid.

The objective of this project is to solve the Navier-Stokes equations, transformed into the streamfunction and vorticity equations, for a two-dimensional lid-driven cavity problem using the finite-difference method. Several iterative solution techniques are considered, including the Gauss-Seidel method, Successive Line Over-Relaxation (SLOR), a multigrid method coupled with Gauss-Seidel iterations, and the Alternating Direction Implicit (ADI) method. The computational domain is a square geometry containing two recesses, and the geometry is symmetric.

After convergence is achieved with each method, the convergence time, the number of iterations required for convergence, and the contours of streamfunction, velocity, and vorticity are evaluated and compared.

Keywords: fluid mechanics, lid-driven cavity, incompressible flow, vorticity, iterative methods, Gauss-Seidel, ADI, SLOR, multigrid.

Theoretical Background

2-1 Governing Equations

First, the continuity equation and the momentum equations in the x- and y-directions are written for an incompressible fluid as follows:

Continuity Equation:

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0$$

X momentum Equation:

$$\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}=-\frac{\partial p}{\partial x}+\frac{1}{Re}(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2})$$

Y momentum Equation:

$$\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}=-\frac{\partial p}{\partial y}+\frac{1}{Re}(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2})$$

The flow is also assumed to be steady. Therefore, the time-dependent terms are neglected and the solution does not vary with time:

Continuity Equation:

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0$$

X momentum Equation:

$$u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}=-\frac{\partial p}{\partial x}+\frac{1}{Re}(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2})$$

Y momentum Equation:

$$u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}=-\frac{\partial p}{\partial y}+\frac{1}{Re}(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2})$$

Differentiating the x-momentum equation with respect to y and the y-momentum equation with respect to x gives:

X momentum:

$$\frac{\partial}{\partial y}\left(u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}=-\frac{\partial p}{\partial x}+\frac{1}{Re}\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)\right)=$$

$$\left(u\frac{\partial^2 u}{\partial x\partial y}+\frac{\partial^2 u}{\partial x\partial y}\right)+\left(v\frac{\partial^2 u}{\partial y^2}+\frac{\partial v}{\partial y}\frac{\partial u}{\partial y}\right)=-\frac{\partial^2 p}{\partial x\partial y}+\frac{1}{Re}\left(\frac{\partial^3 u}{\partial x^2\partial y}+\frac{\partial^3 u}{\partial y^3}\right)$$

Y momentum:

$$\frac{\partial}{\partial x}\left(u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}=-\frac{\partial p}{\partial y}+\frac{1}{Re}\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}\right)\right)=$$

$$\left(\frac{\partial u}{\partial x}\frac{\partial v}{\partial x}+u\frac{\partial^2 v}{\partial x^2}\right)+\left(\frac{\partial^2 v}{\partial x\partial y}+v\frac{\partial^2 v}{\partial x\partial y}\right)=-\frac{\partial^2 p}{\partial x\partial y}+\frac{1}{Re}\left(\frac{\partial^3 v}{\partial x^3}+\frac{\partial^3 v}{\partial y^2\partial x}\right)$$

To eliminate the pressure term, the two differentiated momentum equations are subtracted from one another, yielding:

$$u\frac{\partial}{\partial x}\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right)+v\frac{\partial}{\partial y}\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right)+\left(\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}\right)\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right)=\frac{1}{Re}\left(\frac{\partial^2}{\partial x^2}\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right)+\frac{\partial^2}{\partial y^2}\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right)\right)$$

At this stage, the vorticity and streamfunction formulations are introduced for the velocity components u and v. By definition:

Vorticity Function:

$$\bar{\omega}=\nabla*\bar{V}$$

For a two-dimensional planar flow in the x-y plane:

$$\omega_z=\bar{\omega}.\hat{k}=\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}$$

From the continuity equation, we previously obtained:

$$\frac{\partial v}{\partial y}+\frac{\partial u}{\partial x}=0$$

Using the relations obtained above, the simplified momentum equation can be rewritten as:

$$\begin{cases}\omega_z=\bar{\omega}.\hat{k}=\dfrac{\partial v}{\partial x}-\dfrac{\partial u}{\partial y}\\ \dfrac{\partial v}{\partial y}+\dfrac{\partial u}{\partial x}=0\end{cases}$$

$$=>u\frac{\partial}{\partial x}(\omega_z)+v\frac{\partial}{\partial y}(\omega_z)+(0)(\omega_z)=\frac{1}{Re}\left(\frac{\partial^2}{\partial x^2}(\omega_z)+\frac{\partial^2}{\partial y^2}(\omega_z)\right)$$

$$=>u\frac{\partial\omega_z}{\partial x}+v\frac{\partial\omega_z}{\partial y}=\frac{1}{Re}\left(\frac{\partial^2\omega_z}{\partial x^2}+\frac{\partial^2\omega_z}{\partial y^2}\right)$$

According to the definition of the streamfunction:

$$u=\frac{\partial\psi}{\partial y}$$

$$v=-\frac{\partial\psi}{\partial x}$$

The continuity equation also gives:

$$\frac{\partial v}{\partial y}+\frac{\partial u}{\partial x}=0\rightarrow\frac{\partial^2\psi}{\partial y\partial x}-\frac{\partial^2\psi}{\partial y\partial x}=0$$

Therefore, the continuity equation is automatically satisfied. For ω_z, we also obtain:

$$\omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = -\frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi}{\partial y^2} = -\left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2}\right)$$
$$\rightarrow \ -\bar{\omega} \ = \nabla * \psi$$

At this point, two coupled equations are obtained and must be solved simultaneously:

$$\begin{cases} Poisson\ Equation:\ \omega_z = -\left(\dfrac{\partial^2 \psi}{\partial x^2} + \dfrac{\partial^2 \psi}{\partial y^2}\right) \\ Vorticity\ transport\ Equation: u\dfrac{\partial \omega_z}{\partial x} + v\dfrac{\partial \omega_z}{\partial y} = \dfrac{1}{Re}\left(\dfrac{\partial^2 \omega_z}{\partial x^2} + \dfrac{\partial^2 \omega_z}{\partial y^2}\right) \end{cases}$$

The terms u and v, which have not yet been discretized, must also be discretized using the finite-difference method. These terms are discretized with second-order accuracy on a nonuniform grid. A five-point stencil is used for the second-order discretization. Figure 1 shows these points and the adopted node notation. For simplicity, $\omega_z = \omega$ is used.

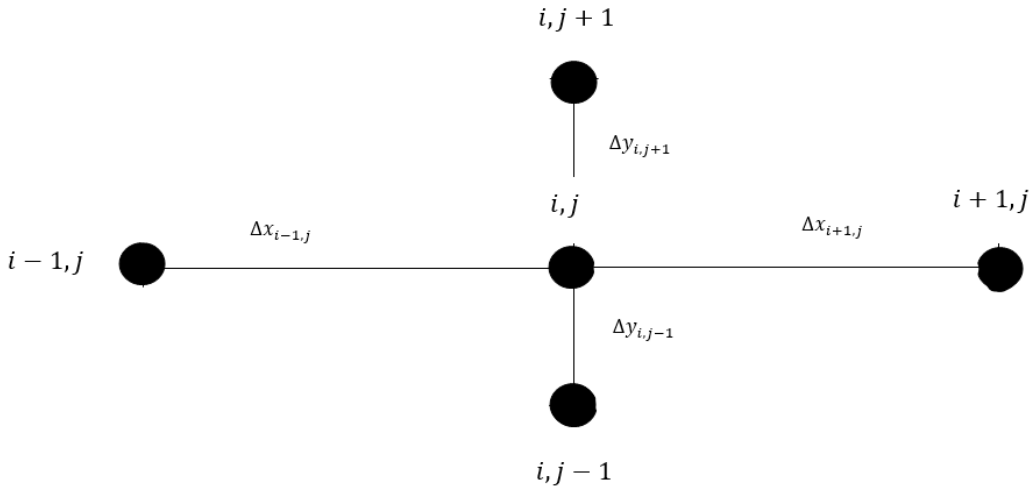


Figure 1: Node arrangement and notation used in the finite-difference method

2-1-1 Discretization of the Poisson Equation

The Poisson equation is discretized as follows:

$$\omega_{i,j} \ = -\frac{\partial^2 \psi}{\partial x^2})_{i,j} - \frac{\partial^2 \psi}{\partial y^2})_{i,j}$$

$$\begin{cases} \left[\psi_{i+1,j} = \psi_{i,j} + \frac{\partial\psi}{\partial x})_{i,j}(\Delta x)_{i+1,j} + \frac{\partial^2\psi}{\partial x^2})_{i,j}\frac{\Delta x^2}{2})_{i+1,j} + O(\Delta x)^3 \quad \rightarrow \times (\Delta x)_{i-1,j}\right] \\ \left[\psi_{i-1,j} = \psi_{i,j} - \frac{\partial\psi}{\partial x})_{i,j}(\Delta x)_{i-1,j} + \frac{\partial^2\psi}{\partial x^2})_{i,j}\frac{\Delta x^2}{2})_{i-1,j} + O(\Delta x)^3 \quad \rightarrow \times (\Delta x)_{i+1,j}\right] \end{cases}$$

$$\Rightarrow \frac{\partial^2\psi}{\partial x^2})_{i,j} = \frac{(\Delta x)_{i-1,j}\psi_{i+1,j} + (\Delta x)_{i+1,j}\psi_{i-1,j} - \psi_{i,j}((\Delta x)_{i-1,j} + (\Delta x)_{i+1,j})}{\frac{1}{2}\left[\left((\Delta x)_{i+1,j}\right)^2(\Delta x)_{i-1,j} + (\Delta x)_{i-1,j}{}^2(\Delta x)_{i+1,j}\right]}$$

$$\Rightarrow \frac{\partial^2\psi}{\partial x^2})_{i,j} = \frac{2\psi_{i+1,j}}{(\Delta x)_{i+1,j}((\Delta x)_{i+1,j} + (\Delta x)_{i-1,j})} + \frac{2\psi_{i-1,j}}{(\Delta x)_{i-1,j}((\Delta x)_{i+1,j} + (\Delta x)_{i-1,j})} - \frac{2\psi_{i,j}}{(\Delta x)_{i-1,j}(\Delta x)_{i+1,j}}$$

Similarly, for $\frac{\partial^2\psi}{\partial y^2})_{i,j}$, we have:

$$\frac{\partial^2\psi}{\partial y^2})_{i,j} = \frac{2\psi_{i,j+1}}{(\Delta y)_{i,j+1}((\Delta y)_{i,j+1} + (\Delta xy_{i,j-1})} + \frac{2\psi_{i,j-1}}{(\Delta y)_{i,j-1}((\Delta y)_{i,j+1} + (\Delta y)_{i,j-1})} - \frac{2\psi_{i,j}}{(\Delta y)_{i,j-1}(\Delta y)_{i,j+1}}$$

Therefore, the Poisson equation becomes:

Poisson Equation:

$$\omega_{i,j} = -\left[\frac{2\psi_{i+1,j}}{(\Delta x)_{i+1,j}((\Delta x)_{i+1,j} + (\Delta x)_{i-1,j})} + \frac{2\psi_{i-1,j}}{(\Delta x)_{i-1,j}((\Delta x)_{i+1,j} + (\Delta x)_{i-1,j})} - \frac{2\psi_{i,j}}{(\Delta x)_{i-1,j}(\Delta x)_{i+1,j}}\right] - \left[\frac{2\psi_{i,j+1}}{(\Delta y)_{i,j+1}((\Delta y)_{i,j+1} + (\Delta y_{i,j-1})} + \frac{2\psi_{i,j-1}}{(\Delta y)_{i,j-1}((\Delta y)_{i,j+1} + (\Delta y)_{i,j-1})} - \frac{2\psi_{i,j}}{(\Delta y)_{i,j-1}(\Delta y)_{i,j+1}}\right]$$

2-1-2 Discretization of the Vorticity Transport Equation

At this stage, the vorticity transport equation must be discretized. The central-difference scheme is used for this discretization.

$$u\frac{\partial\omega_z}{\partial x} + v\frac{\partial\omega_z}{\partial y} = \frac{1}{Re}\left(\frac{\partial^2\omega_z}{\partial x^2} + \frac{\partial^2\omega_z}{\partial y^2}\right)$$

$$\begin{cases} u = \dfrac{\partial \psi}{\partial y} \\ v = -\dfrac{\partial \psi}{\partial x} \end{cases} \rightarrow \frac{\partial \psi}{\partial y}\frac{\partial \omega}{\partial x} - \frac{\partial \psi}{\partial x}\frac{\partial \omega}{\partial y} = \frac{1}{Re}\left(\frac{\partial^2 \omega}{\partial x^2} + \frac{\partial^2 \omega}{\partial y^2}\right)$$

The first-order derivative terms are also discretized using the central-difference scheme:

$$\begin{cases} \psi_{i+1,j} = \psi_{i,j} + \dfrac{\partial \psi}{\partial x})_{i,j}(\Delta x)_{i+1,j} + O(\Delta x^2) \\ \psi_{i-1,j} = \psi_{i,j} - \dfrac{\partial \psi}{\partial x})_{i,j}(\Delta x)_{i-1,j} + O(\Delta x^2) \end{cases}$$

$$\Rightarrow \psi_{i+1,j} - \psi_{i-1,j} = \frac{\partial \psi}{\partial x})_{i,j}((\Delta x)_{i+1,j} + (\Delta x)_{i-1,j}) \Rightarrow \frac{\partial \psi}{\partial x})_{i,j} = \frac{\psi_{i+1,j} - \psi_{i-1,j}}{((\Delta x)_{i+1,j} + (\Delta x)_{i-1,j})}$$

The terms $\frac{\partial \omega}{\partial y}$, $\frac{\partial \omega}{\partial x}$, and $\frac{\partial \psi}{\partial y}$ are discretized using the same procedure, and the following definition is introduced to simplify the equations:

$$\begin{cases} \Delta x^* = ((\Delta x)_{i+1,j} + (\Delta x)_{i-1,j}) \\ \Delta y^* = ((\Delta y)_{i,j+1} + (\Delta y)_{i,j-1}) \end{cases}$$

Finally, the vorticity transport equation takes the following form:

Vorticity Transport Equation:

$$\left(\frac{\psi_{i,j+1} - \psi_{i,j-1}}{\Delta y^*}\right)\left(\frac{\omega_{i+1,j} - \omega_{i-1,j}}{\Delta x^*}\right) - \left(\frac{\psi_{i+1,j} - \psi_{i-1,j}}{\Delta x^*}\right)\left(\frac{\omega_{i,j+1} - \omega_{i,j-1}}{\Delta y^*}\right) =$$
$$\frac{1}{Re}\left[\frac{2\omega_{i+1,j}}{(\Delta x)_{i+1,j}\Delta x^*} + \frac{2\omega_{i-1,j}}{(\Delta x)_{i-1,j}\Delta x^*} - \frac{2\omega_{i,j}}{(\Delta x)_{i+1,j}(\Delta x)_{i,j-1}}\right]$$
$$+ \frac{1}{Re}\left[\frac{2\omega_{i,j+1}}{(\Delta y)_{i,j+1}\Delta y^*} + \frac{2\omega_{i,j-1}}{(\Delta y)_{i,j-1}\Delta y^*} - \frac{2\omega_{i,j}}{(\Delta y)_{i,j+1}(\Delta y)_{i,j-1}}\right]$$

2-2 System Boundary Conditions

Based on the boundary conditions specified for the problem, the boundary conditions for u and v are written. At this stage, the boundary conditions are discretized with first-order accuracy.

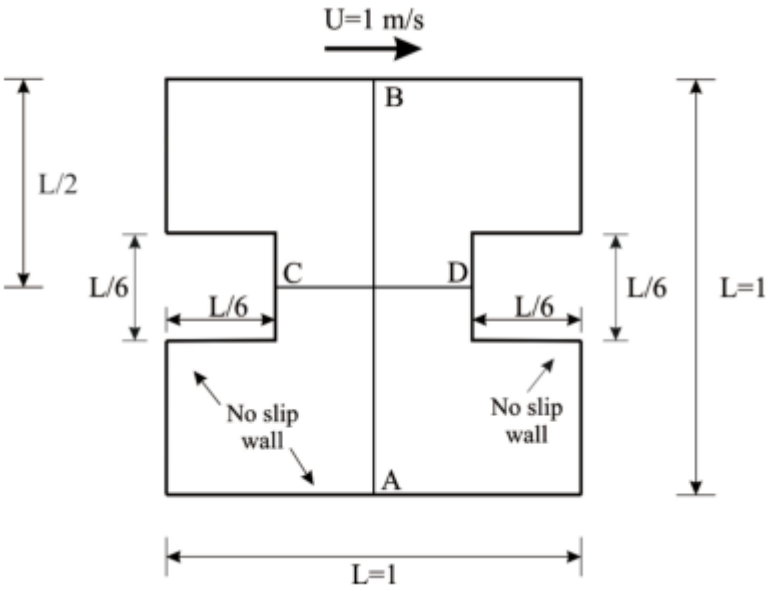


Figure 2: Geometry used to determine the boundary-condition equations

Since the no-slip condition applies at the solid boundaries, the boundary conditions are written as follows:

Bottom Boundary:

$$\begin{cases} u_{i,1} = 0 \\ v_{i,1} = 0 \end{cases}$$

Top Boundary:

$$\begin{cases} u_{i,j_{max}} = 1 \\ v_{i,j_{max}} = 0 \end{cases}$$

Left Boundary:

$$\begin{cases} u_{1,j} = 0 \\ v_{1,j} = 0 \end{cases}$$

Right Boundary:

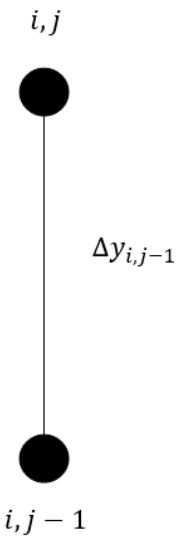
$$\begin{cases} u_{i_{max},j} = 0 \\ v_{i_{max},j} = 0 \end{cases}$$

2-2-1 Modification of Boundary-Condition Relations for ψ, ω

Based on the above relations, the boundary-condition relations for ω, ψ must also be written. At this stage, the central-difference discretization is applied with first-order boundary accuracy. Figures 3 and 4 show the node notation used for this discretization.

2-2-1-1 Boundary Equations for the Upper Walls:

It should be noted that two different cases occur at the upper boundary. According to Figure 2, on the upper wall when $j = L$, $u = 1$; whereas for $j = \frac{5}{12}L$ and $j = \frac{7}{12}L$, $u = 0$. This distinction must be taken into account when discretizing the upper-wall boundary conditions. The nodes used for the discretization are shown in Figure 3:



Boundary conditions for $j = \frac{5}{12}L$ and $j = \frac{7}{12}L$:

Because u and v are zero at the lower boundary, ψ is constant there. For simplicity, this constant is set equal to zero.

Top Boundary:

$$\begin{cases} u = \dfrac{\partial \psi}{\partial y} \\ v = -\dfrac{\partial \psi}{\partial x} \end{cases} \rightarrow u = 1, v = 0 \rightarrow \psi = 0$$

$$\omega_{i,j} = -\left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2}\right) \rightarrow v = 0 \implies \omega_{i,j_{max}} = -\frac{\partial^2 \psi}{\partial y^2})_{i,j}$$

$$\psi_{i,j-1} = \psi_{i,j_{max}} - \Delta y_{i,j-1}\frac{\partial \psi}{\partial y})_{i,j} + \frac{\left(\Delta y_{i,j-1}\right)^2}{2}\frac{\partial^2 \psi}{\partial y^2})_{i,j_{max}} + O(\Delta y)^3$$

$$\implies -\frac{\partial^2 \psi}{\partial y^2})_{i,j_{max}} = \frac{\psi_{i,j} - \psi_{i,j} + \Delta y_{i,j-1}\frac{\partial \psi}{\partial y})_{i,j}}{\frac{\left(\Delta y_{i,j-1}\right)^2}{2}} + O(\Delta y) \xrightarrow{\frac{\partial \psi}{\partial y})_{i,j} = u}$$

$$if\ j = \frac{5}{12}L\ or\ \frac{7}{12}L \rightarrow u = 0 \implies \omega_{i,j_{max}} = -\frac{(\psi_{i,j-1} - \psi_{i,j})}{\frac{(\Delta y_{i,j-1})^2}{2}}$$

Boundary condition for the case j = L:

The procedure is similar to the previous case. The difference between the two cases arises from the u-velocity term. In this case, u = L.

$$\omega_{i,j} = -\left(\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2}\right) \rightarrow v = 0 \implies \omega_{i,j} = -\frac{\partial^2\psi}{\partial y^2})_{i,j}$$

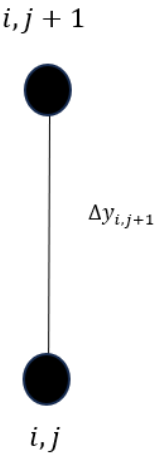
$$\psi_{i,j-1} = \psi_{i,j} - \Delta y_{i,j-1}\frac{\partial\psi}{\partial y})_{i,j} + \frac{(\Delta y_{i,j-1})^2}{2}\frac{\partial^2\psi}{\partial y^2})_{i,j} + O(\Delta y)^3$$

$$\implies -\frac{\partial^2\psi}{\partial y^2})_{i,j} = \frac{\psi_{i,j} - \psi_{i,j} + \Delta y_{i,j-1}\frac{\partial\psi}{\partial y})_{i,j}}{\frac{(\Delta y_{i,j-1})^2}{2}} + O(\Delta y) \xrightarrow{\frac{\partial\psi}{\partial y})_{i,j} = u}$$

$$if\ j = L \rightarrow u = 1 \implies \omega_{i,j} = -\frac{(\psi_{i,j-1} - \psi_{i,j})}{\frac{(\Delta y_{i,j-1})^2}{2}} - \frac{2u}{\Delta y_{i,j-1}} + O(\Delta y)$$

2-2-1-2 Boundary Equations for the Lower Walls:

The node notation used to discretize the boundary condition at the lower wall is shown in Figure 4.



Bottom Boundary:

As for the upper wall, because u and v are zero at the lower boundary, ψ is constant. For simplicity, this constant is set equal to zero.

$$\begin{cases} u = \dfrac{\partial \psi}{\partial y} \\ v = -\dfrac{\partial \psi}{\partial x} \end{cases} \rightarrow u = 0, v = 0 \rightarrow \psi = 0$$

It can also be concluded that ψ is constant in the x-direction and varies only in the y-direction; therefore, variations of ψ occur only in y.
Therefore, we may write:

$$\omega_{i,j} = -\left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2}\right) \rightarrow v = 0 \implies \omega_{i,j} = -\frac{\partial^2 \psi}{\partial y^2})_{i,j}$$

$$\psi_{i,j+1} = \psi_{i,j} + \Delta y_{i,j-1} \frac{\partial \psi}{\partial y})_{i,j} + \frac{\left(\Delta y_{i,j+1}\right)^2}{2} \frac{\partial^2 \psi}{\partial y^2})_{i,j} + O(\Delta y)^3$$

$$\implies -\frac{\partial^2 \psi}{\partial y^2})_{i,j} = \frac{\psi_{i,j} - \psi_{i,j} - \Delta y_{i,j+1} \frac{\partial \psi}{\partial y})_{i,j}}{\frac{\left(\Delta y_{i,j+1}\right)^2}{2}} + O(\Delta y) \xrightarrow{\frac{\partial \psi}{\partial y})_{i,j} = u}$$

$$\rightarrow u = 0 \implies \omega_{i,j} = \frac{-(\psi_{i,j+1} - \psi_{i,j})}{\frac{\left(\Delta y_{i,j+1}\right)^2}{2}}$$

2-2-1-3 Boundary Equations for the Right Walls:

The following node notation is used to discretize the boundary equations on the right side:

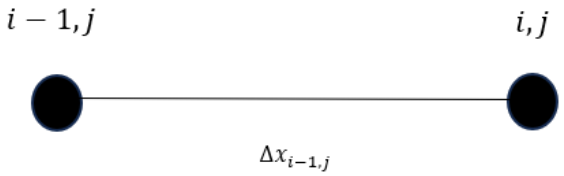


Figure 5: Node notation used for discretization in the x-direction at the right wall

According to the problem geometry, the right-wall boundary conditions can be written for the two cases $i = L$ and $i = \frac{2L}{3}$:

Right Boundary:

$$\begin{cases} u = \dfrac{\partial \psi}{\partial y} \\ v = -\dfrac{\partial \psi}{\partial x} \end{cases} \rightarrow u = 0, v = 0 \rightarrow \psi = 0$$

As on the left boundary, ψ is constant in the y-direction. Therefore:

$$\omega_{i,j} = -\left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2}\right) = -\frac{\partial^2 \psi}{\partial x^2})_{i,j}$$

$$\psi_{i-1,j} = \psi_{i,j} - \Delta x_{i-1,j} \frac{\partial \psi}{\partial x})_{i,j} + \frac{\left(\Delta x_{i-1,j}\right)^2}{2} \frac{\partial^2 \psi}{\partial x^2})_{i,j} + O(\Delta x)^3$$

$$=> \frac{\partial^2 \psi}{\partial x^2})_{i,j} = -\frac{\psi_{i,j} - \psi_{i-1,j}}{\frac{\left(\Delta x_{i-1,j}\right)^2}{2}} + \frac{\Delta x_{i-1,j} \frac{\partial \psi}{\partial x})_{i,j}}{\frac{\left(\Delta x_{i-1,j} x\right)^2}{2}} + O(\Delta x)$$

$$\begin{cases} \omega_{i,j} = -\dfrac{\partial^2 \psi}{\partial x^2})_{i,j} \\ v = -\dfrac{\partial \psi}{\partial x} \end{cases} \rightarrow \omega_{i,j} = \frac{\psi_{i-1,j} - \psi_{i,j}}{\frac{\left(\Delta x_{i-1,j}\right)^2}{2}} + \frac{-\Delta x_{i-1,j} \frac{\partial \psi}{\partial x})_{i,j}}{\frac{\left(\Delta x_{i-1,j}\right)^2}{2}} + O(\Delta x) \rightarrow v = 0$$

$$\rightarrow \omega_{i,j} = \frac{2(\psi_{i-1,j} - \psi_{i,j})}{\left(\Delta x_{i-1,j}\right)^2} + O(\Delta x)$$

2-2-1-4 Boundary Equations for the Left Walls:

Left Boundary:

$$\begin{cases} u = \dfrac{\partial \psi}{\partial y} \\ v = -\dfrac{\partial \psi}{\partial x} \end{cases} \rightarrow u = 0, v = 0 \rightarrow \psi = 0$$

In this case, ψ is constant in the y-direction. Therefore:

$$\omega_{i,j} = -\left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2}\right) = -\frac{\partial^2 \psi}{\partial x^2})_{i,j}$$

$$\psi_{i+1,j} = \psi_{i,j} + \Delta x_{i+1,j} \frac{\partial \psi}{\partial x})_{i,j} + \frac{\left(\Delta x_{i+1,j}\right)^2}{2} \frac{\partial^2 \psi}{\partial x^2})_{i,j} + O(\Delta x)^3$$

$$=> \frac{\partial^2 \psi}{\partial x^2})_{i,j} = -\frac{\psi_{i,j} - \psi_{i+1,j}}{\frac{\left(\Delta x_{i+1,j}\right)^2}{2}} + \frac{\Delta x_{i+1,j} \frac{\partial \psi}{\partial x})_{i,j}}{\frac{\left(\Delta x_{i+1,j} x\right)^2}{2}} + O(\Delta x)$$

$$\begin{cases} \omega_{i,j} = -\dfrac{\partial^2 \psi}{\partial x^2})_{i,j} \\ v = -\dfrac{\partial \psi}{\partial x} \end{cases} \rightarrow \omega_{i,j} = \frac{\psi_{i+1,j} - \psi_{i,j}}{\frac{\left(\Delta x_{i+1,j}\right)^2}{2}} + \frac{-\Delta x_{i+1,j} \frac{\partial \psi}{\partial x})_{i,j}}{\frac{\left(\Delta x_{i+1,j}\right)^2}{2}} + O(\Delta x) \rightarrow v = 0$$

$$\rightarrow \omega_{i,j} = \frac{2(\psi_{i+1,j} - \psi_{i,j})}{\left(\Delta x_{i+1,j}\right)^2} + O(\Delta x)$$

2-3 Theoretical Background of the Gauss-Seidel Method

One approach for solving a system of algebraic equations is to use an iterative method until convergence is achieved. The Gauss-Seidel method is one such technique. Diagonal dominance is a sufficient condition for convergence; that is, the magnitude of each diagonal coefficient should be greater than the contributions of the remaining coefficients in the corresponding row. Because this is only a sufficient condition, convergence may still occur even when it is not satisfied. In this method, the equations are rearranged into explicit update relations for the unknowns. An initial guess is assigned to all unknowns, after which the iterations continue until the convergence criterion is satisfied. During each sweep, the most recently updated values are used whenever they are available on the right-hand side of the equations.

This section describes the derivation of the equations used in the Gauss-Seidel method. The adopted node notation and the selected nodes are shown below. Nodes with known values are indicated in red.

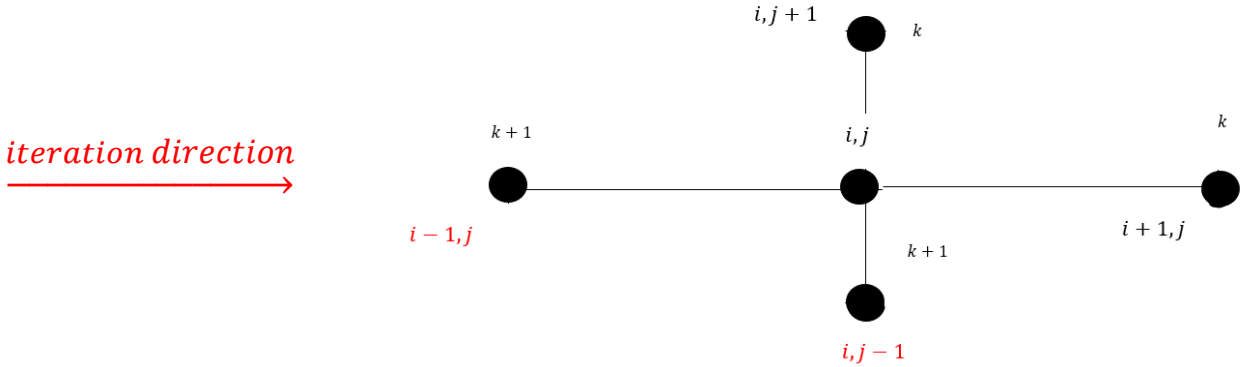


Figure 6: Selected nodes and notation used to derive the equations

After defining the nodes, the Gauss-Seidel solution formulation is obtained using the equations derived in Sections 1-1-2 and 2-1-2.

2-3-1 Poisson Equation:

The Poisson equation is considered first.

$$\omega_{i,j}^{k} = \left[\frac{2\psi_{i+1,j}^{k}}{(\Delta x)_{i+1,j}((\Delta x)_{i+1,j} + (\Delta x)_{i-1,j})} + \frac{2\psi_{i-1,j}^{k+1}}{(\Delta x)_{i-1,j}((\Delta x)_{i+1,j} + (\Delta x)_{i-1,j})} - \frac{2\,\psi_{i,j}^{k+1}}{(\Delta x)_{i-1,j}(\Delta x)_{i+1,j}}\right] - \left[\frac{2\psi_{i,j+1}^{k}}{(\Delta y)_{i,j+1}((\Delta y)_{i,j+1} + (\Delta y_{i,j-1})} + \frac{2\psi_{i,j-1}^{k+1}}{(\Delta y)_{i,j-1}((\Delta y)_{i,j+1} + (\Delta y)_{i,j-1})} - \frac{2\,\psi_{i,j}^{k+1}}{(\Delta y)_{i,j-1}(\Delta y)_{i,j+1}}\right]$$

Solving the Poisson equation provides $\psi_{i,j}^{k+1}$. Therefore, the equation is rearranged to obtain $\psi_{i,j}^{k+1}$:

$$\psi_{i,j}^{k+1}\left[\frac{2}{(\Delta x)_{i+1,j}(\Delta x)_{i-1,j}}+\frac{2}{(\Delta y)_{i,j+1}\Delta y_{i,j-1}}\right]$$
$$=\omega_{i,j}{}^{k}+\frac{2\,\psi_{i+1,j}^{k}}{(\Delta x)_{i+1,j}\Delta x^{*}}+\frac{2\psi_{i-1,j}^{k+1}}{(\Delta x)_{i-1,j}\Delta x^{*}}+\frac{2\psi_{i,j+1}^{k}}{(\Delta y)_{i,j+1}\Delta y^{*}}+\frac{2\psi_{i,j-1}^{k+1}}{(\Delta y)_{i,j-1}\Delta y^{*}}$$

Thus, the update equation for $\psi_{i,j}^{k+1}$ becomes:

$$\psi_{i,j}^{k+1}=\left[\frac{\omega_{i,j}{}^{k}+\frac{2\,\psi_{i+1,j}^{k}}{(\Delta x)_{i+1,j}\Delta x^{*}}+\frac{2\psi_{i-1,j}^{k+1}}{(\Delta x)_{i-1,j}\Delta x^{*}}+\frac{2\psi_{i,j+1}^{k}}{(\Delta y)_{i,j+1}\Delta y^{*}}+\frac{2\psi_{i,j-1}^{k+1}}{(\Delta y)_{i,j-1}\Delta y^{*}}}{\left[\frac{2}{(\Delta x)_{i+1,j}(\Delta x)_{i-1,j}}+\frac{2}{(\Delta y)_{i,j+1}\Delta y_{i,j-1}}\right]}\right]$$

This is the first equation used in the Gauss-Seidel solution procedure. Once $\psi_{i,j}^{k+1}$ has been obtained, the boundary equations can be updated and the result can also be used to evaluate the second governing equation.

2-3-2 Vorticity Transport Equation

As stated in Section 1-3-2, the vorticity transport equation is examined in this section:

$$\left(\frac{\psi_{i,j+1}^{k}-\psi_{i,j-1}^{k+1}}{\Delta y^{*}}\right)\left(\frac{\omega_{i+1,j}^{k}-\omega_{i-1,j}^{k+1}}{\Delta x^{*}}\right)-\left(\frac{\psi_{i+1,j}^{k}-\psi_{i-1,j}^{k+1}}{\Delta x^{*}}\right)\left(\frac{\omega_{i,j+1}^{k}-\omega_{i,j-1}^{k+1}}{\Delta y^{*}}\right)=$$
$$\frac{1}{Re}\left[\frac{2\omega_{i+1,j}^{k}}{(\Delta x)_{i+1,j}\Delta x^{*}}+\frac{2\omega_{i-1,j}^{k+1}}{(\Delta x)_{i-1,j}\Delta x^{*}}-\frac{2\omega_{i,j}^{k+1}}{(\Delta x)_{i+1,j}(\Delta x)_{i,j-1}}\right]$$
$$+\frac{1}{Re}\left[\frac{2\omega_{i,j+1}^{k}}{(\Delta y)_{i,j+1}\Delta y^{*}}+\frac{2\omega_{i,j-1}^{k+1}}{(\Delta y)_{i,j-1}\Delta y^{*}}-\frac{2\omega_{i,j}^{k+1}}{(\Delta y)_{i,j+1}(\Delta y)_{i,j-1}}\right]$$

Rearranging the equations to calculate $\omega_{i,j}^{k+1}$ gives:

$$\omega_{i,j}^{k+1}\left[-\left[\frac{2}{(\Delta x)_{i+1,j}(\Delta x)_{i-1,j}}+\frac{2}{(\Delta y)_{i,j+1}\Delta y_{i,j-1}}\right]\right]$$
$$=Re\left[\left(\frac{\psi_{i,j+1}^{k}-\psi_{i,j-1}^{k+1}}{\Delta y^{*}}\right)\left(\frac{\omega_{i+1,j}^{k}-\omega_{i-1,j}^{k+1}}{\Delta x^{*}}\right)-\left(\frac{\psi_{i+1,j}^{k}-\psi_{i-1,j}^{k+1}}{\Delta x^{*}}\right)\left(\frac{\omega_{i,j+1}^{k}-\omega_{i,j-1}^{k+1}}{\Delta y^{*}}\right)\right]$$
$$-\left[\frac{2\omega_{i+1,j}^{k}}{(\Delta x)_{i+1,j}\Delta x^{*}}+\frac{2\omega_{i-1,j}^{k+1}}{(\Delta x)_{i-1,j}\Delta x^{*}}\right]-\left[\frac{2\omega_{i,j+1}^{k}}{(\Delta y)_{i,j+1}\Delta y^{*}}+\frac{2\omega_{i,j-1}^{k+1}}{(\Delta y)_{i,j-1}\Delta y^{*}}\right]$$

Thus, the update equation for $\omega_{i,j}^{k+1}$ becomes:

$$\omega_{i,j}^{k+1} = \frac{Re\left[\left(\frac{\psi_{i,j+1}^{k}-\psi_{i,j-1}^{k+1}}{\Delta y^*}\right)\left(\frac{\omega_{i+1,j}^{k}-\omega_{i-1,j}^{k+1}}{\Delta x^*}\right)-\left(\frac{\psi_{i+1,j}^{k}-\psi_{i-1,j}^{k+1}}{\Delta x^*}\right)\left(\frac{\omega_{i,j+1}^{k}-\omega_{i,j-1}^{k+1}}{\Delta y^*}\right)\right]}{\left[-\left[\frac{2}{(\Delta x)_{i+1,j}(\Delta x)_{i-1,j}}+\frac{2}{(\Delta y)_{i,j+1}\Delta y_{i,j-1}}\right]\right]}$$
$$+\frac{-\left[\frac{2\omega_{i+1,j}^{k}}{(\Delta x)_{i+1,j}\Delta x^*}+\frac{2\omega_{i-1,j}^{k+1}}{(\Delta x)_{i-1,j}\Delta x^*}\right]-\left[\frac{2\omega_{i,j+1}^{k}}{(\Delta y)_{i,j+1}\Delta y^*}+\frac{2\omega_{i,j-1}^{k+1}}{(\Delta y)_{i,j-1}\Delta y^*}\right]}{\left[-\left[\frac{2}{(\Delta x)_{i+1,j}(\Delta x)_{i-1,j}}+\frac{2}{(\Delta y)_{i,j+1}\Delta y_{i,j-1}}\right]\right]}$$

2-4 Theory of the SLOR Method

The SLOR method is an iterative solution technique in which the unknowns along an entire line are updated during each iteration. For a clearer explanation, the following figure is taken from the third edition of Computational Fluid Mechanics and Heat Transfer by Anderson:

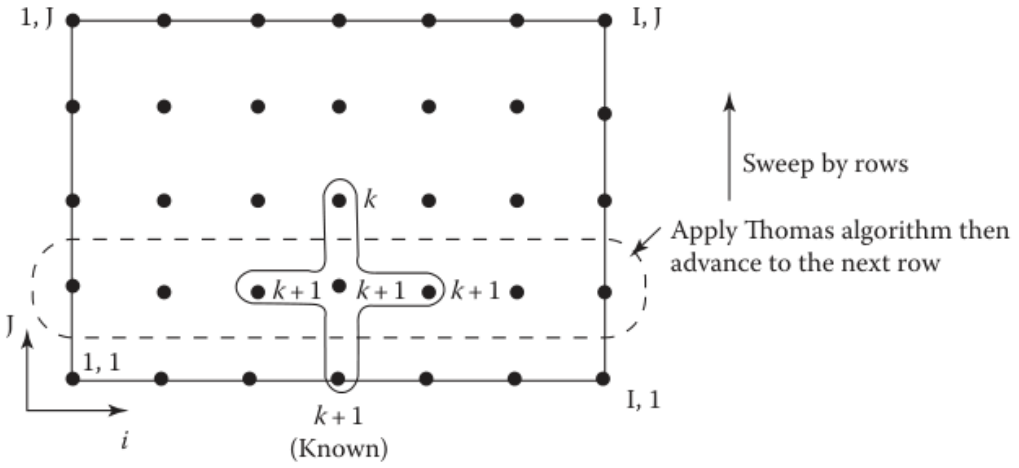


Figure 7: Line SOR method

In this method, the unknown values along the row j = n are first calculated using the Thomas algorithm; the Poisson equation is used for this step. The calculation then proceeds along the columns, where the vorticity transport equation is solved to determine the unknowns in the i = m direction. For example, when computing the solution at point (i,j) on row j = n at iteration k+1, the points (i-1,j), (i,j), and (i+1,j) on that row are updated together. The value at i,j-1 has already been evaluated at iteration k+1 on the previous row, whereas the value associated with the not-yet-updated neighboring row is taken from iteration k. Thus, the line equation contains three unknown terms and three known terms.

After this brief description of the method, the Poisson and vorticity transport equations are rearranged for the SLOR solution procedure. In this section, nodes with known values are shown in red. For deriving the equations, the sweep direction for obtaining the points in row (:,j) of the Poisson equation is taken as $\xrightarrow{\textit{iteration direction}}$, while the sweep direction for obtaining the points along (i,:) in the vorticity transport equation is $\uparrow$ *iteration direction*.

To derive the SLOR equations, the equations are first written in SOR form and the SLOR procedure is then applied.

2-4-1 Equations Based on the SOR Solution Method:

Poisson Equation:

$$\frac{2}{(\Delta x)_{i+1,j}\Delta x^*}\psi_{i+1,j}^{k+1} - \left(\frac{2}{(\Delta x)_{i+1,j}(\Delta x)_{i-1,j}} + \frac{2}{(\Delta y)_{i,j+1}\Delta y_{i,j-1}}\right)\psi_{i,j}^{k+1} + \left(\frac{2}{(\Delta x)_{i+1,j}(\Delta x)_{i-1,j}}\right)\psi_{i-1,j}^{k+1} = -\frac{2\psi_{i,j+1}^{k}}{(\Delta y)_{i,j+1}\Delta y^*} - \frac{2\psi_{i,j-1}^{k+1}}{(\Delta y)_{i,j-1}\Delta y^*} - \omega_{i,j}^{k}$$

$$\theta_1 = \left[\frac{2}{(\Delta x)_{i+1,j}\Delta x^*}\right], \theta_2 = -\left[\frac{2}{(\Delta x)_{i+1,j}(\Delta x)_{i-1,j}} + \frac{2}{(\Delta y)_{i,j+1}\Delta y_{i,j-1}}\right], \theta_3 = \left[\frac{2}{(\Delta x)_{i+1,j}(\Delta x)_{i-1,j}}\right],$$
$$\theta_4 = -\frac{2\psi_{i,j+1}^{k}}{(\Delta y)_{i,j+1}\Delta y^*} - \frac{2\psi_{i,j-1}^{k+1}}{(\Delta y)_{i,j-1}\Delta y^*} - \omega_{i,j}^{k}$$

Vorticity Transport Equation:

$$\omega_{i,j}^{k+1} = -\left(\frac{\psi_{i,j+1}^{k} - \psi_{i,j-1}^{k+1}}{\Delta y^*\Delta x^*} - \frac{2}{Re(\Delta x)_{i-1,j}\Delta x^*}\right) + \left(\frac{2}{(\Delta x)_{i+1,j}\Delta x^*} + \frac{2}{(\Delta y)_{i,j+1}\Delta y^*}\right)\omega_{i,j}^{k+1} + \left(\frac{\psi_{i,j+1}^{k} - \psi_{i,j-1}^{k+1}}{\Delta y^*\Delta x^*} - \frac{2}{Re(\Delta x)_{i-1,j}\Delta x^*}\right)\omega_{i+1,j}^{k+1} = \left(\frac{\psi_{i,j+1}^{k+1} - \psi_{i-1,j}^{k+1}}{\Delta y^*}\right)\left(\frac{\omega_{i,j+1}^{k} - \omega_{i,j-1}^{k+1}}{\Delta x^*}\right) + \frac{1}{Re}\left(\frac{2\omega_{i,j+1}^{k}}{(\Delta y)_{i,j+1}\Delta y^*} + \frac{2\omega_{i,j-1}^{k+1}}{(\Delta y)_{i,j-1}\Delta y^*}\right)$$

$$\beta_1 = -\left(\frac{\psi_{i,j+1}^{k} - \psi_{i,j-1}^{k+1}}{\Delta y^*\Delta x^*} - \frac{2}{Re(\Delta x)_{i-1,j}\Delta x^*}\right), \beta_2 = \left(\frac{2}{(\Delta x)_{i+1,j}\Delta x^*} + \frac{2}{(\Delta y)_{i,j+1}\Delta y^*}\right),$$
$$\beta_3 = \left[\left(\frac{\psi_{i,j+1}^{k} - \psi_{i,j-1}^{k+1}}{\Delta y^*\Delta x^*} - \frac{2}{Re(\Delta x)_{i-1,j}\Delta x^*}\right)\right],$$
$$\beta_4 = \left(\frac{\psi_{i,j+1}^{k+1} - \psi_{i-1,j}^{k+1}}{\Delta y^*}\right)\left(\frac{\omega_{i,j+1}^{k} - \omega_{i,j-1}^{k+1}}{\Delta x^*}\right) + \frac{1}{Re}\left(\frac{2\omega_{i,j+1}^{k}}{(\Delta y)_{i,j+1}\Delta y^*} + \frac{2\omega_{i,j-1}^{k+1}}{(\Delta y)_{i,j-1}\Delta y^*}\right)$$

After obtaining the SOR form of the equations, they are rewritten in SLOR form.

2-4-2 Poisson Equation:

The Poisson equation is rearranged in SLOR form:

$$\psi_{i,j}^{k+1} = \lambda\left[-\psi_{i,j}^{k} - \frac{\theta_3}{\theta_2}\psi_{i-1,j}^{k+1} + \frac{\theta_4}{\theta_2} - \frac{\theta_1}{\theta_2}\psi_{i+1,j}^{k+1}\right] + (1-\lambda)\psi_{i,j}^{k}$$

$$a_1 = \lambda\frac{\theta_3}{\theta_2}, \quad a_2 = 1, \quad a_3 = \frac{\theta_1}{\theta_2}\lambda, \quad (RHS)_{i,j_1} = (1-\lambda)\psi_{i,j}^{k}$$

2-4-3 Vorticity Transport Equation

The vorticity transport equation is considered in this section:

$$\omega_{i,j}^{k+1} == \lambda\left[-\frac{\beta_1}{\beta_2}\omega_{i-1,j}^{k+1} - \frac{\beta_3}{\beta_2}\omega_{i+1,j}^{k+1} + \frac{\beta_4}{\beta_2} - \omega_{i,j}^{k}\right] + (1-\lambda)\omega_{i,j}^{k}$$

$$b_1 = \lambda\frac{\beta_1}{\beta_2}, \quad b_2 = 1, \quad b_3 = \frac{\beta_3}{\beta_2}\lambda, \quad (RHS)_{i,j_2} = \lambda\frac{\beta_4}{\beta_2} + (1-\lambda)\omega_{i,j}^{k}$$

Using the coefficients defined for both equations, the following relations are obtained:

$$\begin{cases} a_{1_{i,j}}\psi_{i-1,j}^{k+1} + a_{2_{i,j}}\psi_{i,j}^{k+1} + a_{3_{i,j}}\psi_{i+1,j}^{k+1} = (RHS)_{i,j_1} \\ b_{1_{i,j}}\omega_{i-1,j}^{k+1} + b_{2_{i,j}}\omega_{i,j}^{k+1} + b_3\,\omega_{i+1,j}^{k+1} = (RHS)_{i,j_2} \end{cases}$$

To simplify the coefficient-matrix assembly procedure, the coefficient matrix is first derived for a 4 x 4 mesh:

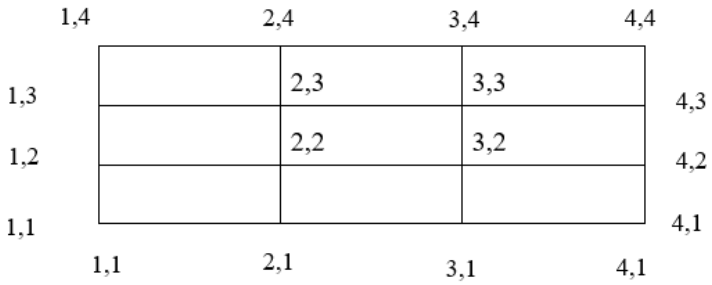


Figure 8: Example of a 4 x 4 mesh

$$i = j = 1 \rightarrow \psi_{1,1} = 0$$

$$i = 2, j = 1 \rightarrow a_{1_{2,1}}\psi_{1,1}^{k+1} + a_{2_{2,1}}\psi_{2,1}^{k+1} + a_{3_{2,1}}\psi_{3,1}^{k+1} = (RHS)_{2,1}$$

$$i = 3, j = 1 \rightarrow a_{1_{3,1}}\psi_{2,1}^{k+1} + a_{2_{3,1}}\psi_{i,j}^{k+1} + a_{3_{3,1}}\psi_{i-1,j}^{k+1} = (RHS)_{3,1}$$

$$i = 4, \quad j = 1 \rightarrow \psi_{4,1} = 0$$

$$i = 1, \quad j = 2 \rightarrow \psi_{1,2} = 0$$

$$i = 2, j = 2 \rightarrow a_{1_{2,2}}\psi_{1,2}^{k+1} + a_{2_{2,2}}\psi_{2,2}^{k+1} + a_{3_{2,2}}\psi_{3,2}^{k+1} = (RHS)_{2,2}$$

$$i = 3, j = 2 \rightarrow a_{1_{3,2}}\psi_{2,2}^{k+1} + a_{2_{3,2}}\psi_{3,2}^{k+1} + a_{3_{3,2}}\psi_{3,2}^{k+1} = (RHS)_{3,2}$$

$$i = 4, \quad j = 2 \rightarrow \psi_{4,2} = 0$$

$$i = 1, \quad j = 3 \rightarrow \psi_{2,3} = 0$$

$$i = 2, j = 3 \rightarrow a_{1_{2,3}}\psi_{1,3}^{k+1} + a_{2_{2,3}}\psi_{2,3}^{k+1} + a_{3_{2,3}}\psi_{3,3}^{k+1} = (RHS)_{2,3}$$

$$i=3, j=3 \rightarrow a_{1_{3,3}}\psi_{2,3}^{k+1} + a_{2_{3,3}}\psi_{3,3}^{k+1} + a_{3_{3,3}}\psi_{4,3}^{k+1} = (RHS)_{3,3}$$
$$i=4, \quad j=3 \rightarrow \psi_{4,3}=0$$
$$i=1, \quad j=4 \rightarrow \psi_{1,4}=0$$
$$i=2, j=4 \rightarrow a_{1_{2,4}}\psi_{1,4}^{k+1} + a_{2_{2,4}}\psi_{2,4}^{k+1} + a_{3_{2,4}}\psi_{3,4}^{k+1} = (RHS)_{2,4}$$
$$i=3, j=4 \rightarrow a_{1_{3,4}}\psi_{2,4}^{k+1} + a_{2_{3,4}}\psi_{3,4}^{k+1} + a_{3_{3,4}}\psi_{4,4}^{k+1} = (RHS)_{3,4}$$
$$i=4, \quad j=4 \rightarrow \psi_{4,4}=0$$

For the 4 x 4 mesh, the coefficient matrix is assembled as follows:

Coefficient Matrix										Unknown Vector	Known-Value Vector
1	0	0	0	0	0	0	0	...	0	$\psi_{1,1}$	0
a_1	a_2	a_3	0	0	0	0	0	...	0	$\psi_{2,1}$	$(RHS)_{2,1}$
0	a_1	a_2	a_3	0	0	0	0	...	0	$\psi_{3,1}$	$(RHS)_{3,1}$
0	0	0	1	0	0	0	0	...	0	$\psi_{4,1}$	⋮
0	0	0	0	1	0	0	0	...	0	$\psi_{1,2}$	⋮
0	0	0	0	a_1	a_2	a_3	0	...	0	$\psi_{2,2}$	⋮
0	0	0	0	0	a_1	a_2	a_3	...	0	$\psi_{3,2}$	⋮
0	0	0	0	0	0	0	1	...	0	$\psi_{4,2}$	⋮
⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	$\psi_{1,3}$	⋮
⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	$\psi_{2,3}$	⋮
⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	$\psi_{3,3}$	⋮
⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	$\psi_{4,3}$	⋮
⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	$\psi_{1,4}$	⋮
0	0	0	...	...	0	a_1	a_2	a_3	0	$\psi_{2,4}$	$(RHS)_{2,4}$
0	0	0	...	...	0	0	a_1	a_2	a_3	$\psi_{3,4}$	$(RHS)_{3,4}$
0	0	0	...	...	0	0	0	0	1	$\psi_{4,4}$	0

Following the same procedure, the coefficient matrix for the Poisson equation is obtained for an n x n mesh.

As stated above, the coefficients obtained for the Poisson equation are as follows:

$$a_{1_{i,j}}\psi_{i-1,j}^{k+1} + a_{2_{i,j}}\psi_{i,j}^{k+1} + a_{3_{i,j}}\psi_{i+1,j}^{k+1} = (RHS)_{i,j_1}$$

The coefficient matrix therefore has the following form:

Coefficient Matrix										Unknown Vector	Known-Value Vector
1	0	0	0	0	0	0	0	...	0	$\psi_{1,1}$	0
$(a_1)_{2,1}$	$(a_2)_{2,1}$	$(a_3)_{2,1}$	0	0	0	0	0	...	0	$\psi_{2,1}$	$(RHS)_{2,1}$
0	$(a_1)_{3,1}$	$(a_2)_{3,1}$	$(a_3)_{3,1}$	0	0	0	0	...	0	$\psi_{3,1}$	$(RHS)_{3,1}$
⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	...	0	⋮	⋮
⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	...	...	⋮	⋮
⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	...	...	⋮	⋮
⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	...	...	⋮	⋮
⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	...	...	⋮	⋮
⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋮	⋮
⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋮	⋮
⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋮	⋮
⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋮	⋮
⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	⋱	$\psi_{I-3,1}$	$(RHS)_{I-3,1}$
0	0	0	...	...	0	0	0	0	0	$\psi_{I-2,1}$	$(RHS)_{I-2,1}$
0	0	0	...	...	0	0	$(a_1)_{I-1,1}$	$(a_2)_{I-1,1}$	$(a_3)_{I-1,1}$	$\psi_{I-1,1}$	$(RHS)_{I-1,1}$
0	0	0	...	...	0	0	0	0	1	$\psi_{I,1}$	0

$$if\ i = 1, j = 1 \ \rightarrow \ (RHS)_{1,1} = 0$$
$$if\ i = I, j = 1 \ \rightarrow \ (RHS)_{I,1} = 0$$

The coefficient matrix for the vorticity transport equation is assembled in the same manner.

2-5 ADI Method

2-6 Multigrid Method

The objective of the multigrid method is to eliminate low-frequency error components on the fine grid. Although a fine grid is appropriate for obtaining an accurate converged solution, low-frequency errors decay slowly and therefore reduce the convergence rate. By transferring the problem to a coarser grid, these low-frequency errors on the fine grid appear as higher-frequency errors and can be reduced more efficiently. Consequently, convergence can be achieved with fewer iterations. In practice, the error residual is transferred from the fine grid to a coarse grid by restriction, the error equation is solved on the coarse grid, and the resulting correction is transferred back to the fine grid by prolongation. Thus, the multigrid (MG) procedure provides an improved initial estimate for subsequent Gauss-Seidel iterations and reduces the number of Gauss-Seidel iterations required for convergence.

In the first step, ten Gauss-Seidel iterations are performed. The error matrix is then calculated on the fine grid and transferred to the coarse grid. The injection method is used for this fine-to-coarse transfer. Injection is the simplest restriction procedure: boundary points are transferred directly without modification, and the remaining interior points are sampled alternately from the fine grid. This approach is applicable to uniform grids; for nonuniform grids, weighted averaging should be used. The transfer procedure is illustrated schematically in the following figure:

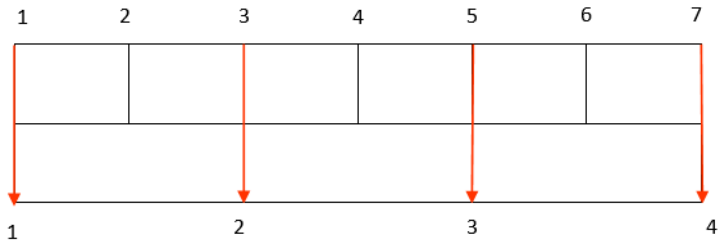


Figure 9: Illustration of the injection method

In the second step, the correction equation is solved on the coarser grid using a zero initial guess, and the iterations continue until convergence is reached. The resulting correction is then transferred to the finer grid using the prolongation process.

In the third step, the correction is added to the original solution and the result is used as a new initial guess for the Gauss-Seidel solution on the fine grid.

The transfer between fine and coarse grids and the correction of the initial estimate are repeated until convergence is achieved. Convergence is reached when successive fine-grid Gauss-Seidel solutions no longer exhibit a significant difference. A schematic of the two-level multigrid method is shown below:

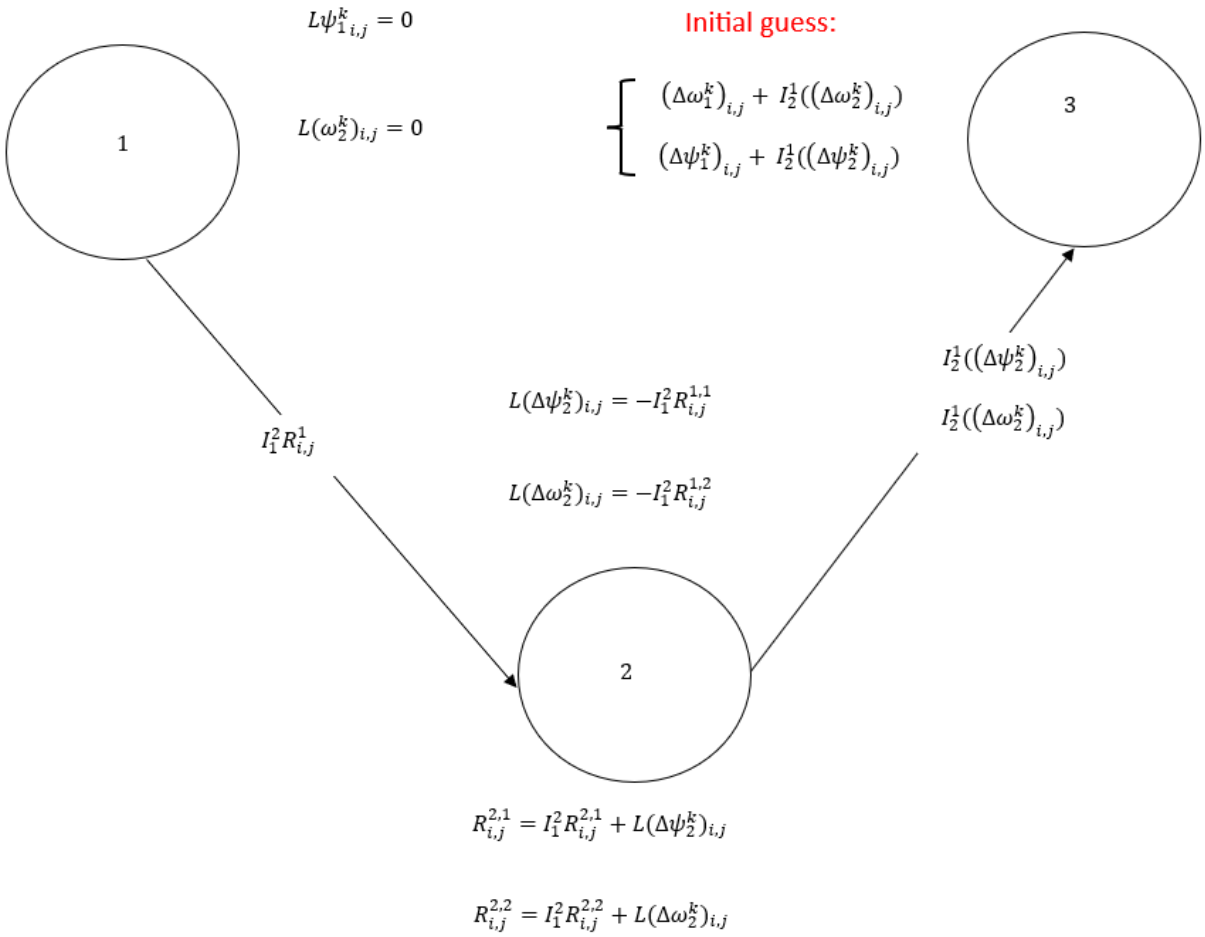


Figure 10: Schematic of the multigrid method

Code Description (Geometry Construction and Mesh Generation)

To construct the geometry considered in this project, its boundaries are divided into several segments. Each segment is labeled as shown in the following figure.

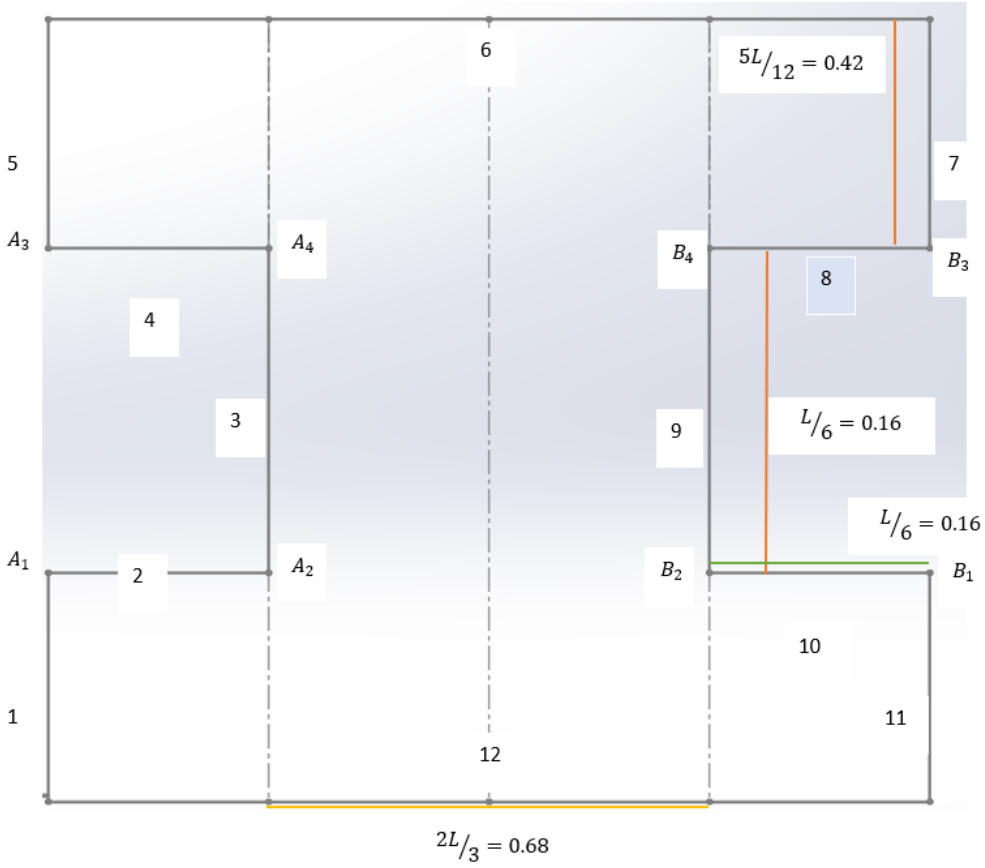


Figure 11: Grid-segment notation used for mesh generation

According to the labels and dimensions shown in Figure 11, the geometry is meshed using the relation $n = {}^{l}/_{h}$. To avoid mesh-generation errors, the values ${}^{L}/_{6}$, ${}^{2L}/_{3}$, and ${}^{5L}/_{12}$ are selected as indicated in the figure.

After mesh generation, the grid points are organized using the defined mesh_sorting function so that the computational grid conforms to the required geometry.

For implementation, the geometry is divided into three regions:

1- *The region defined by* $i = 1:{}^{L}/_{h}$ *and* $j = 1:{}^{0.42L}/_{h} - 1$ *is considered the first geometric region.*

2- *The region defined by* $i = {}^{0.16L}/_{h}:\frac{(0.16+0.68)L}{h}$ *and* $j = {}^{0.42L}/_{h}:\frac{(0.42+0.16)L}{h}$ *is considered the second geometric region.*

3- *The third region is the same as the first region in the i-direction, while in the j-direction it is defined by* $1 + \frac{(0.42+0.16)L}{h}$.

Results

This section presents the results obtained using each of the requested solution methods.

4-1 Gauss-Seidel Method

The results obtained using the Gauss-Seidel method are presented below.

4-1-1 Task 1

The plots of y versus u along line AB for Reynolds numbers of 1, 10, and 100 are shown below:

4-1-1-1 u profile along line AB at Re = 1:

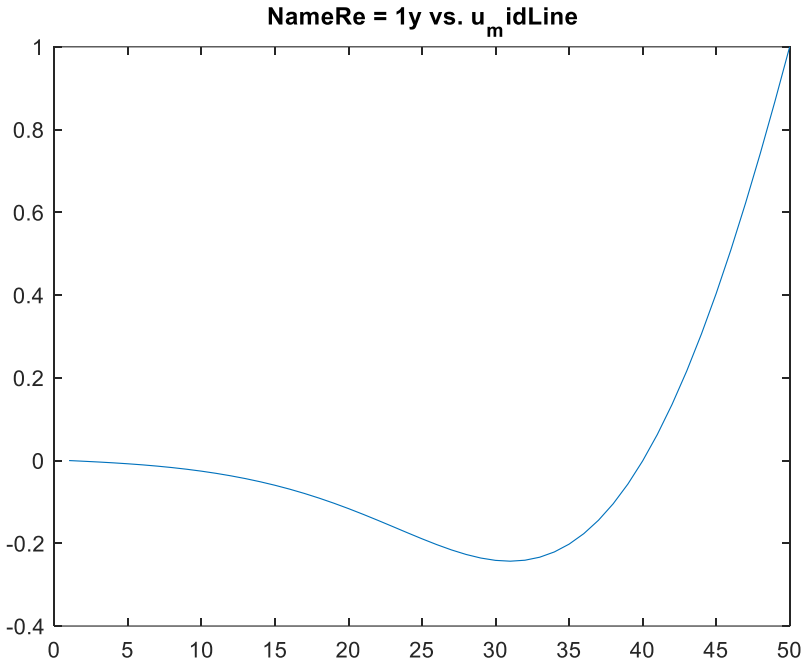


Figure 12: y versus u_{AB} at Re = 1

4-1-1-2 u profile along line AB at Re = 10:

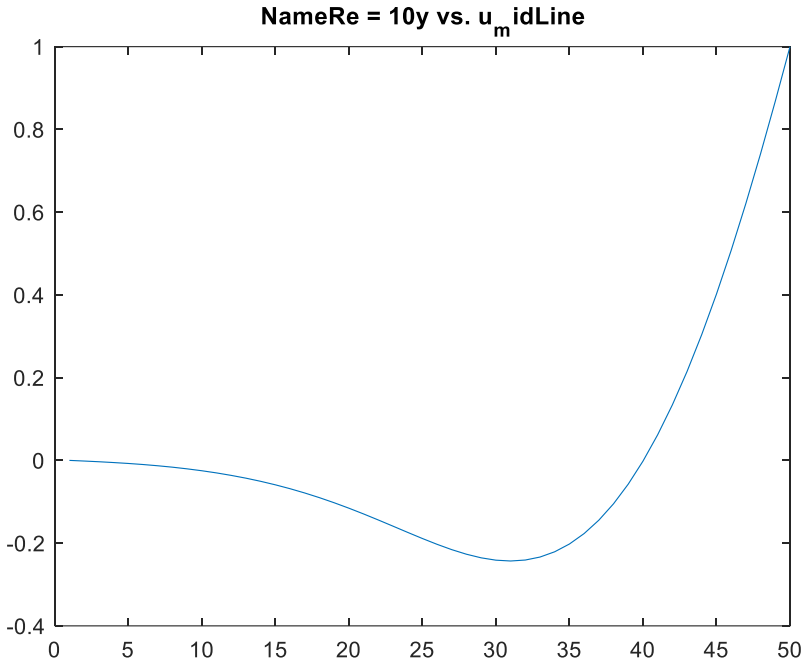


Figure 13: y versus u_{AB} at Re = 10

4-1-1-3 u profile along line AB at Re = 100:

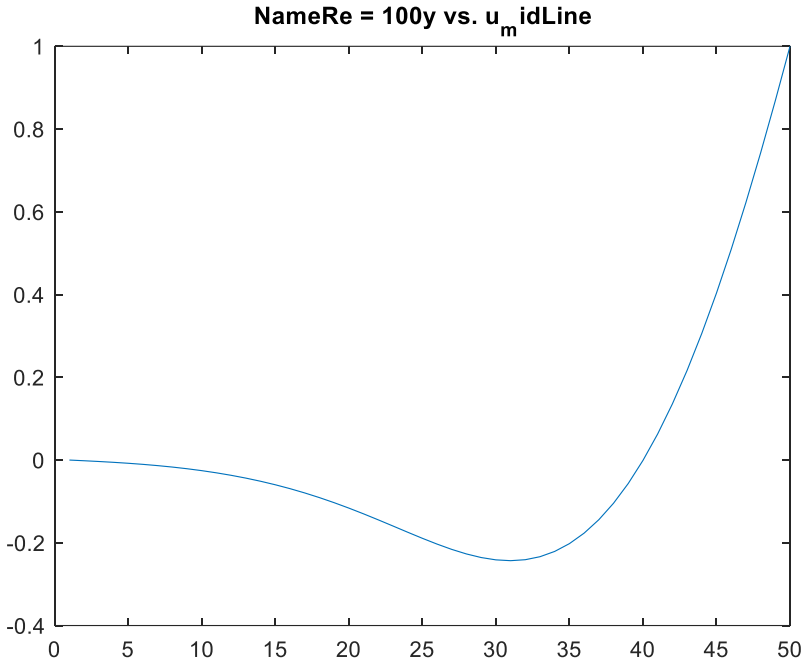


Figure 14: y versus u_{AB} at Re = 100

4-1-1-4 Plot of v versus x along line CD at Re = 1:

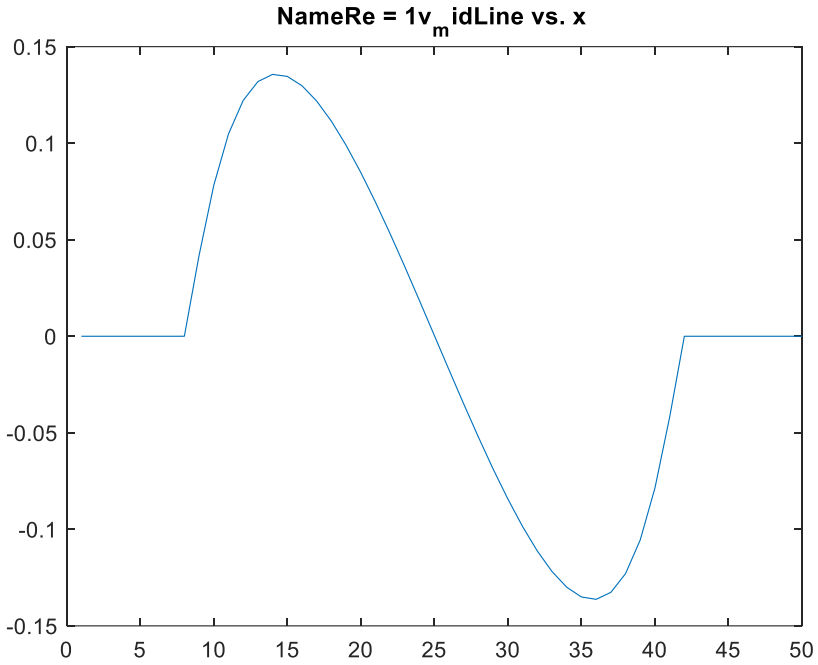


Figure 15: v_{CD} versus x at Re = 1

4-1-1-5 Plot of v versus x along line CD at Re = 10:

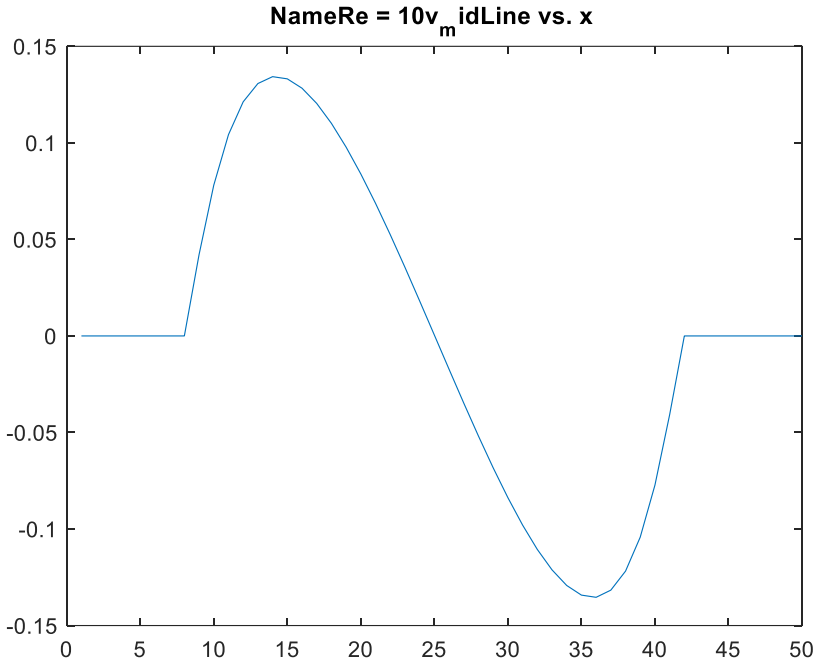


Figure 16: v_{CD} versus x at Re = 10

4-1-1-6 Plot of v versus x along line CD at Re = 100:

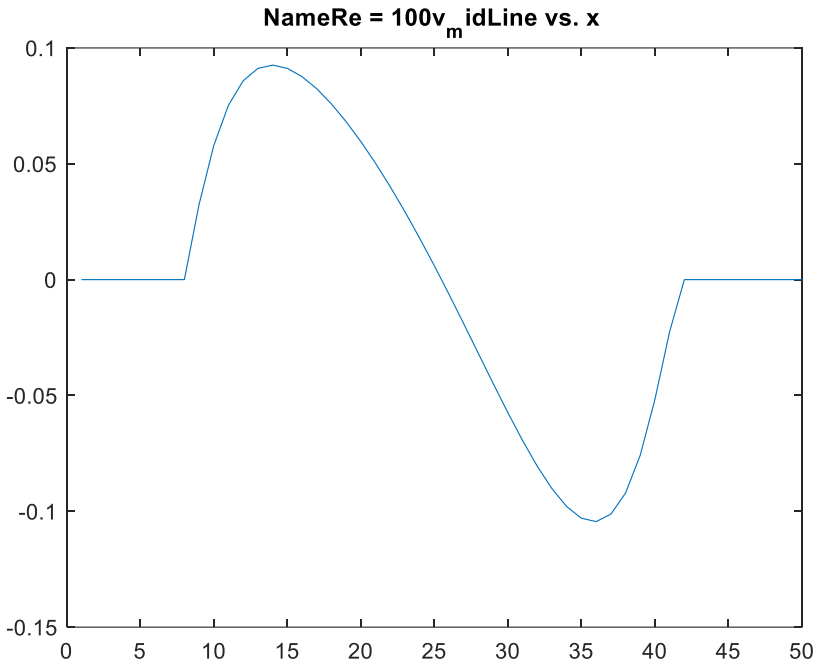


Figure 16: v_{CD} versus x at Re = 100

4-1-2 Task 2:

Contours of ψ, ω, u, and v at Re = 1:

4-1-2-1 ψ contour at Re = 1:

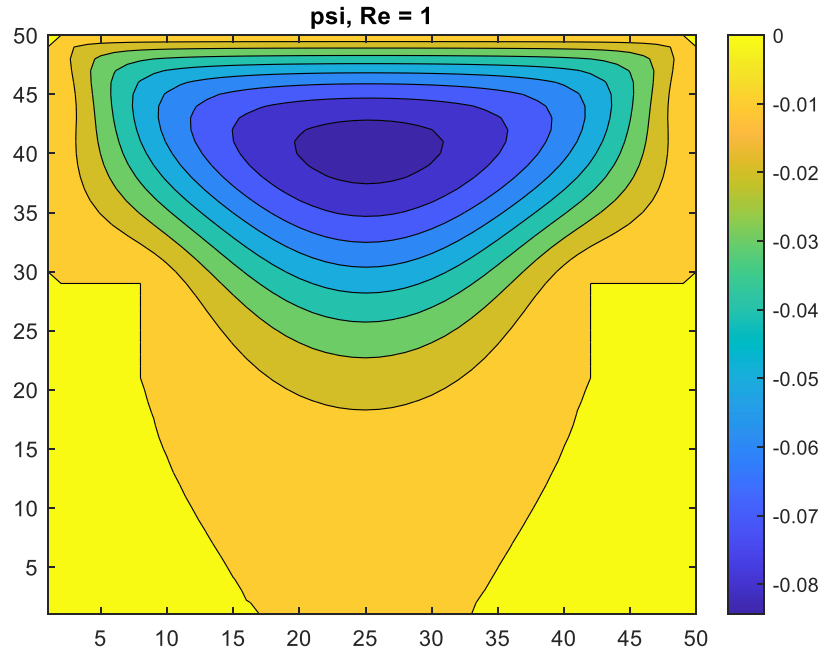


Figure 16: ψ contour at Re = 1

4-1-2-2 ω contour at Re = 1:

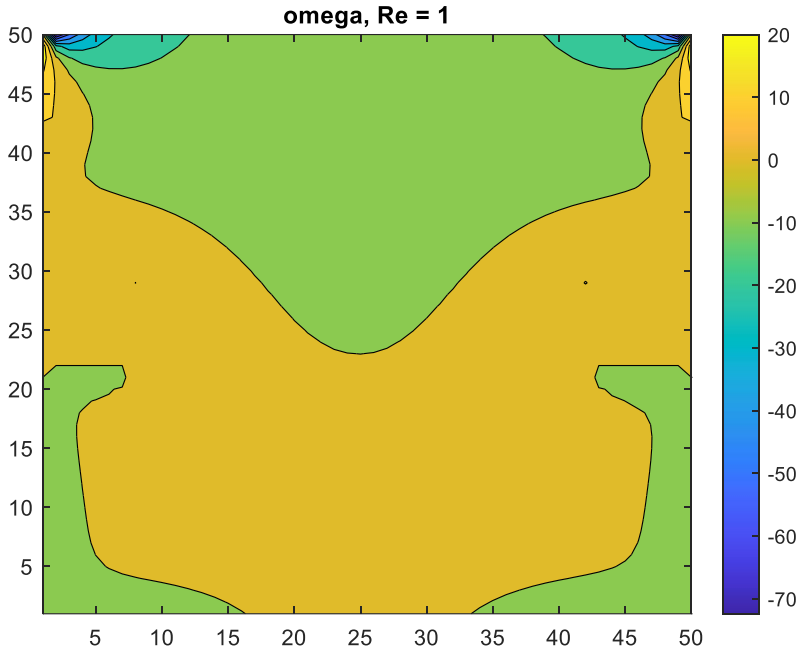


Figure 17: ω contour at Re = 1

4-1-2-3 u contour at Re = 1:

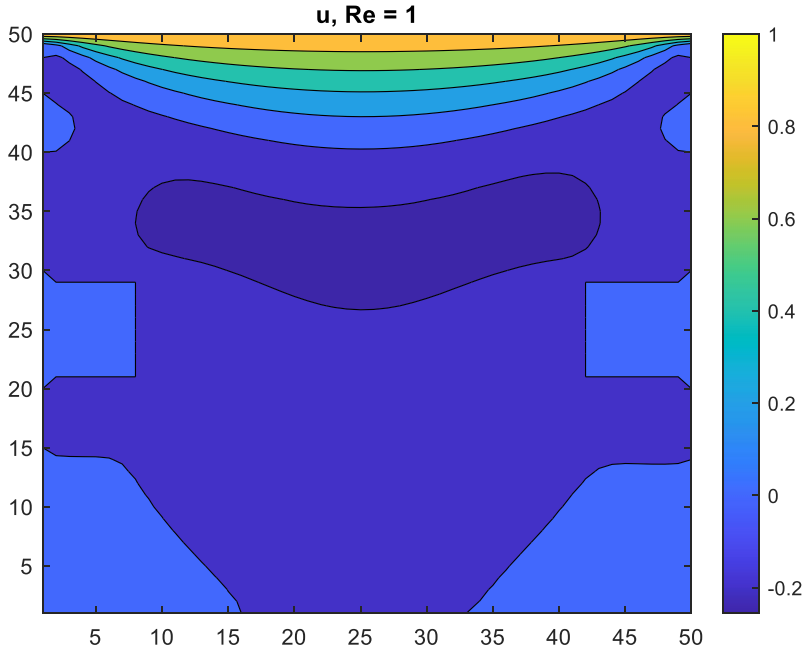


Figure 18: u contour at Re = 1

4-1-2-4 v contour at Re = 1:

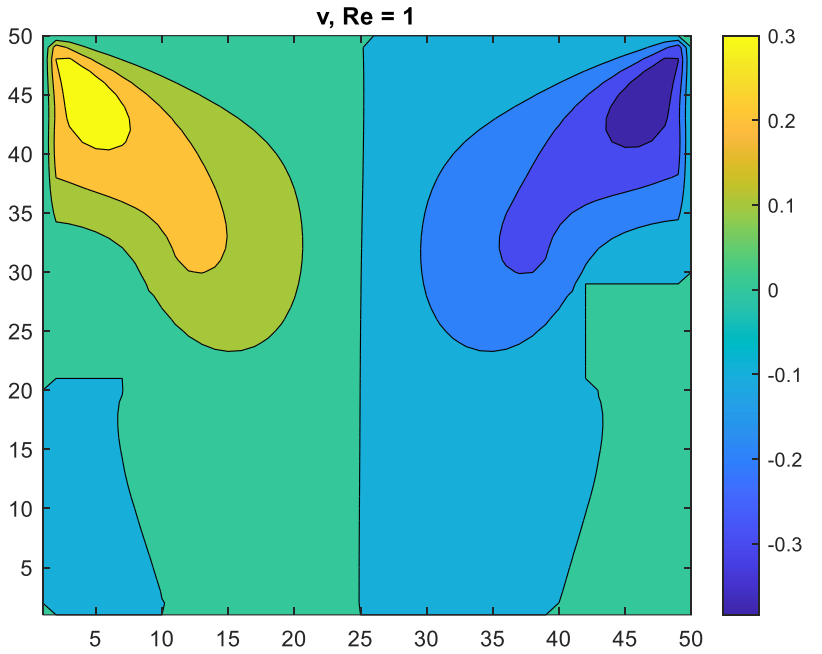


Figure 18: v contour at Re = 1

4-1-3 Contours of ψ, ω, u, and v at Re = 10:

4-1-3-1 ψ contour at Re = 10:

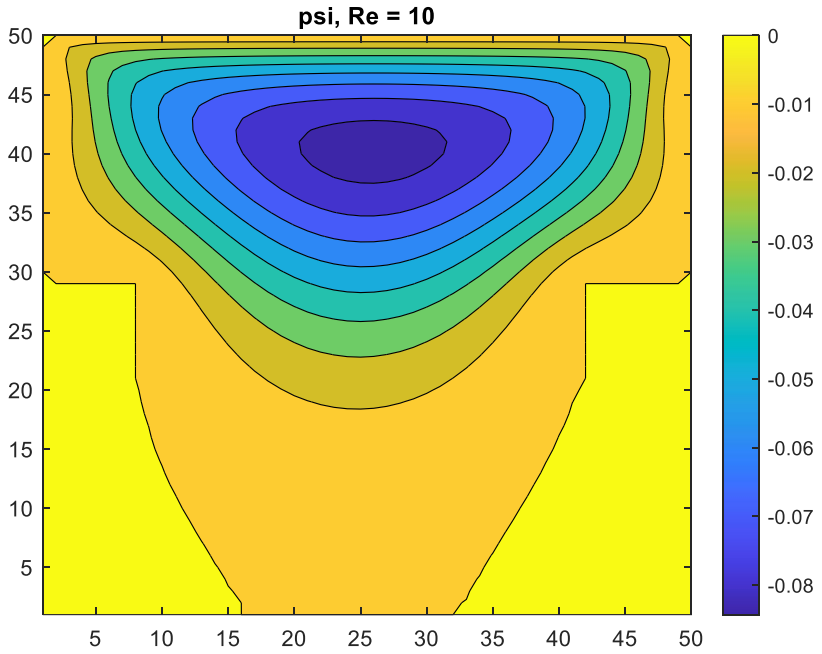


Figure 19: ψ contour at Re = 10

4-1-3-2 ω contour at Re = 10:

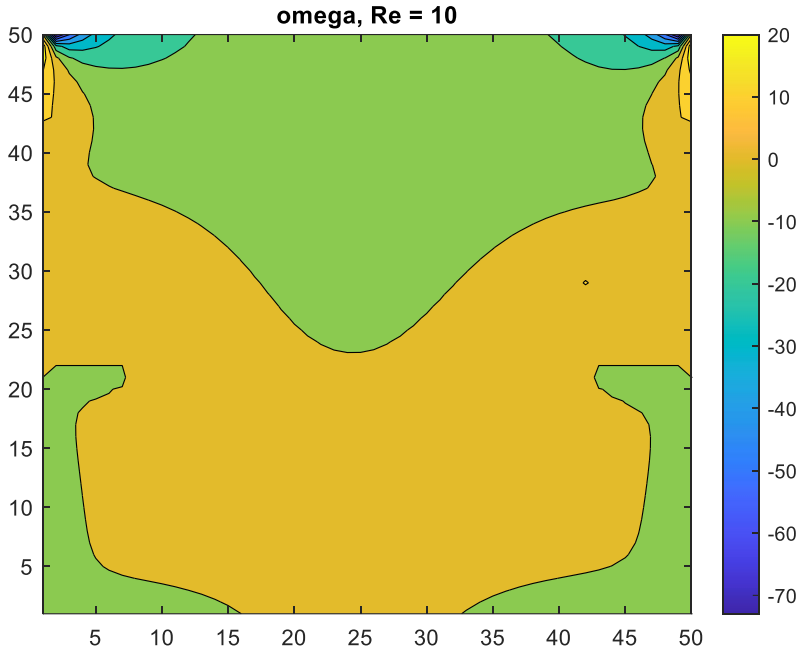


Figure 20: ω contour at Re = 10

4-1-3-3 u contour at Re = 10:

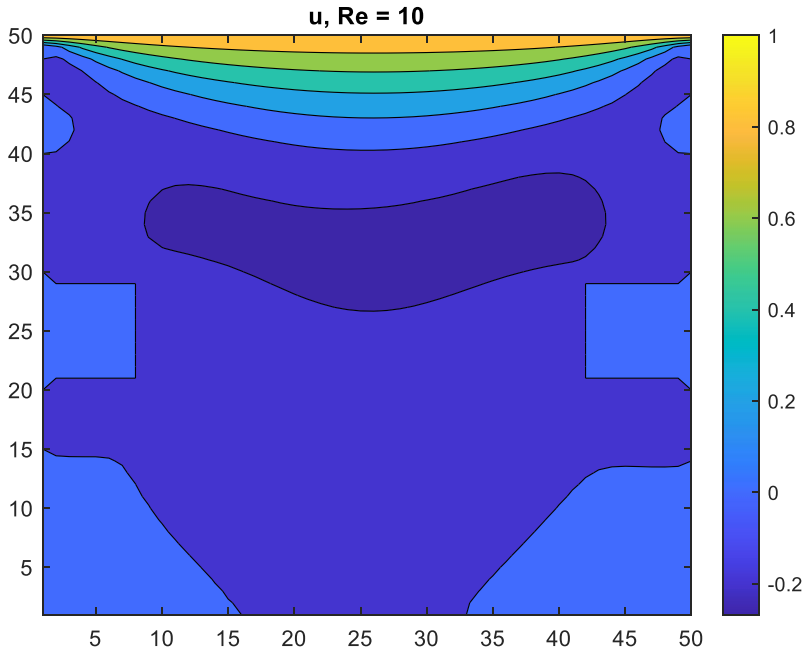


Figure 21: u contour at Re = 10

4-1-3-4 v contour at Re = 10:

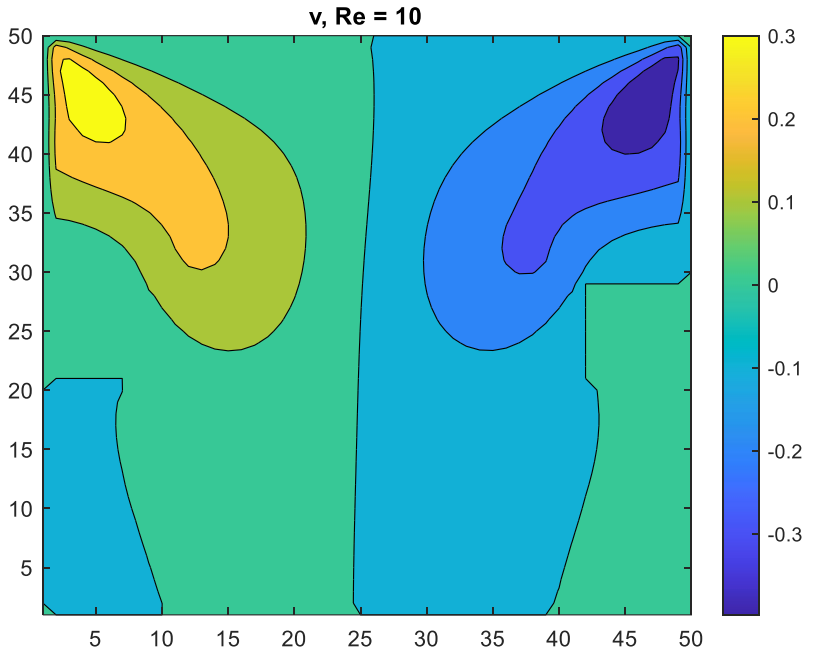


Figure 22: v contour at Re = 10

4-1-4 Contours of ψ, ω, u, and v at Re = 100:

4-1-4-1 ψ contour at Re = 100:

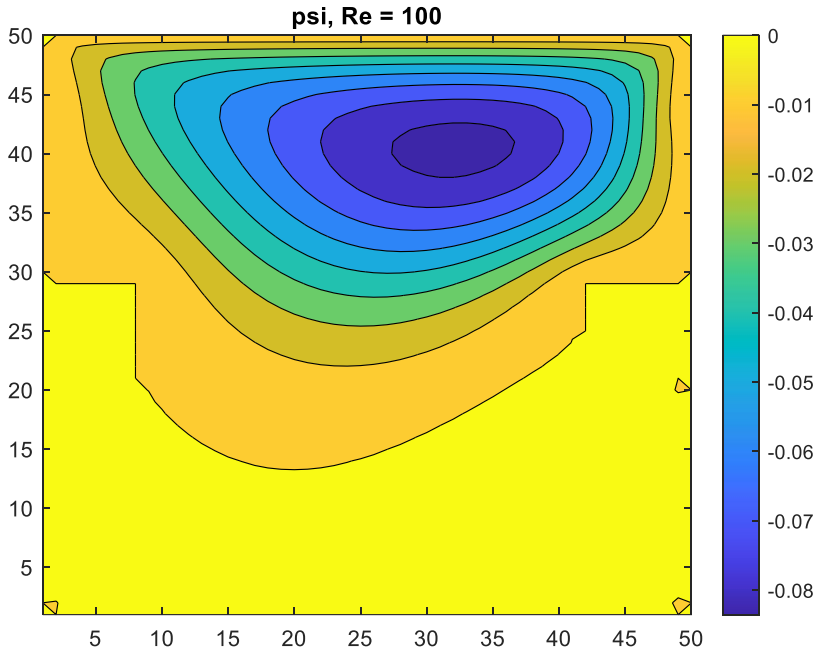


Figure 23: ψ contour at Re = 100

4-1-4-2 ω contour at Re = 100:

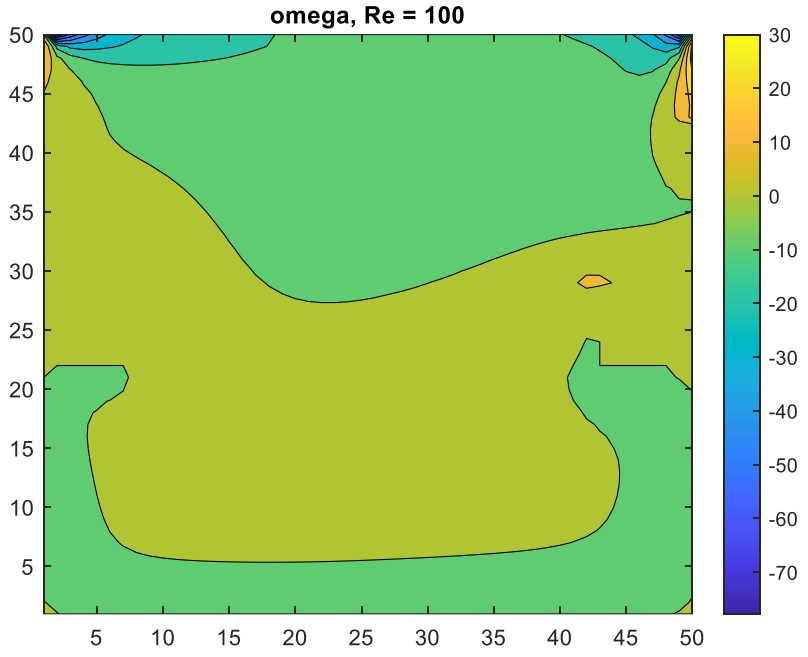


Figure 24: ω contour at Re = 100

4-1-4-3 u contour at Re = 100:

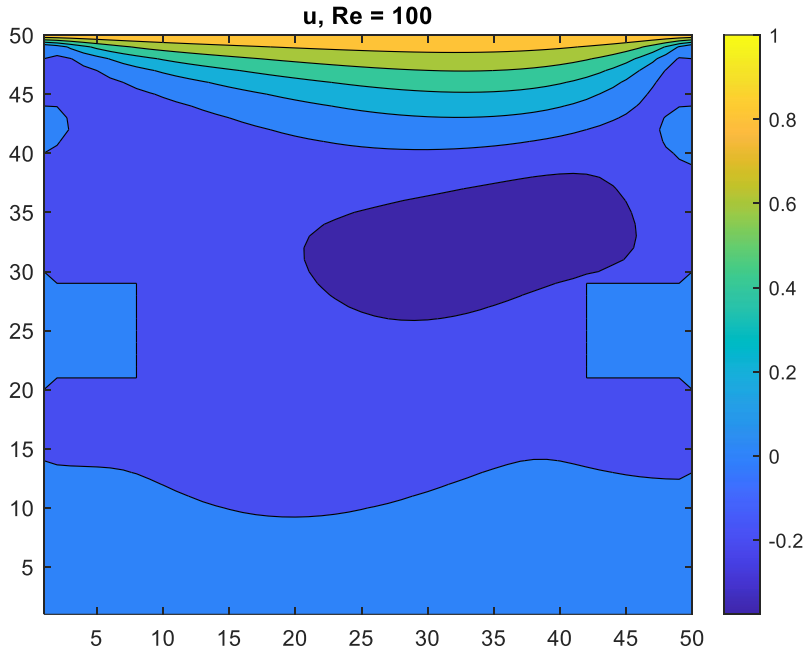


Figure 25: u contour at Re = 100

4-1-4-4 v contour at Re = 100:

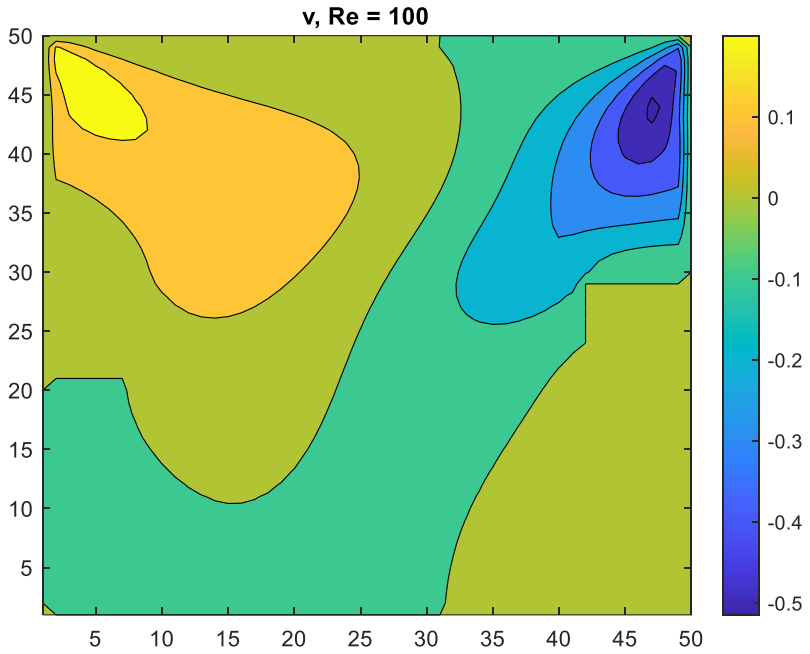


Figure 26: v contour at Re = 100

4-1-5 Task 3:

The solution time is measured using the MATLAB toc command and is displayed in the Command Window at the end of the computation:

Solution times for Re = 1, 10, and 100:

Re	Time (s)
1	4.632
10	4.6981
100	5.8322

4-1-6 Task 4:

Number of iterations required to reach convergence at Re = 1, 10, and 100:

Re	Number of Iterations
1	2086
10	2102
100	2636

4-2 SLOR Results

The results obtained using the SLOR method are presented below.

4-2-1 Task 1

The plots of y versus u along line AB for Re = 1, 10, and 100 are shown below:

The plots of v versus x along line CD for Re = 1, 10, and 100 are shown below:

4-2-2 Task 2

4-2-2-1 ψ contour at Re = 1:

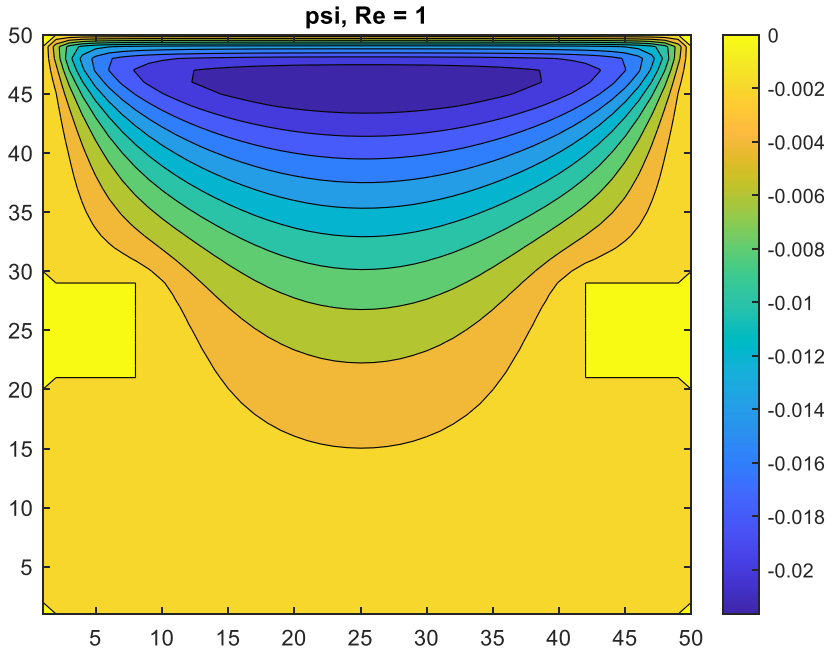


Figure 16: ψ contour at Re = 1

4-2-2-2 ω contour at Re = 1:

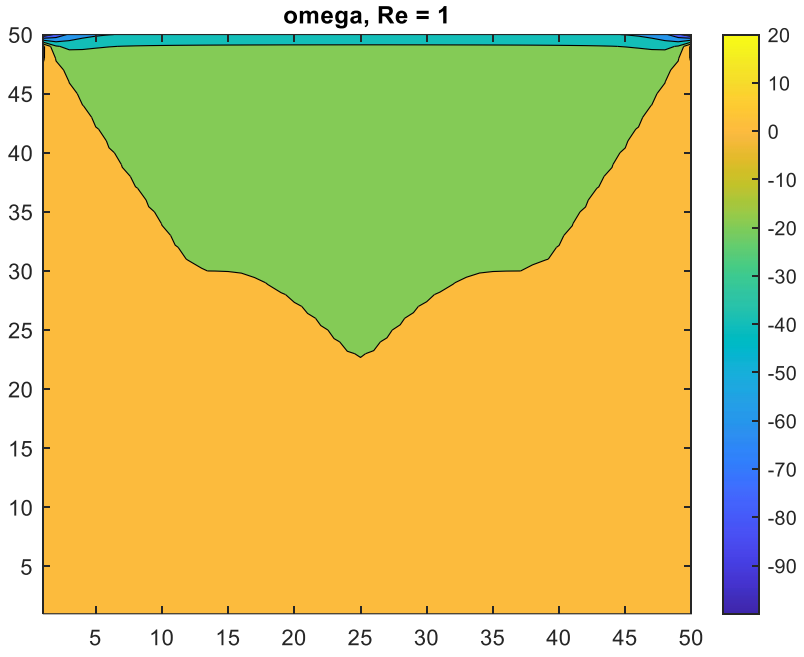


Figure 17: ω contour at Re = 1

4-2-2-3 u contour at Re = 1:

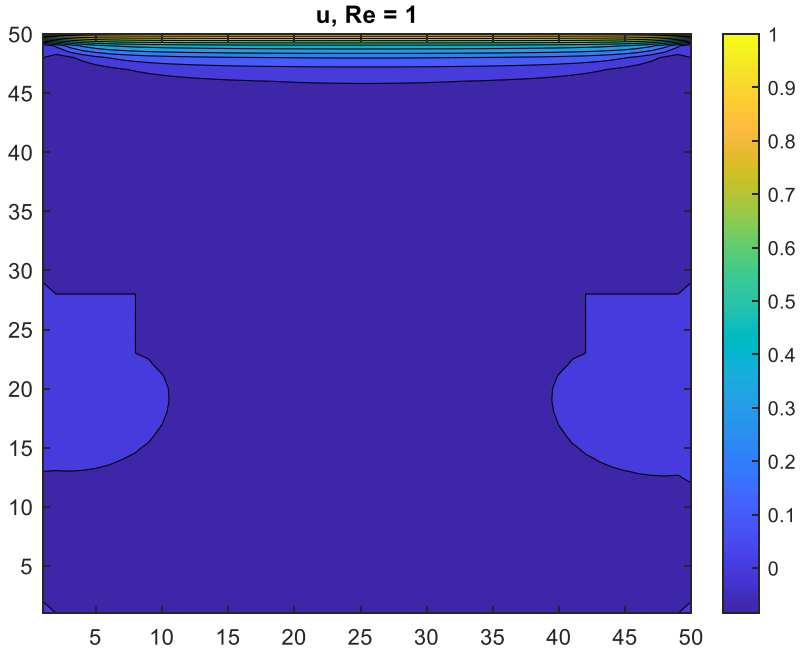


Figure 18: u contour at Re = 1

4-2-2-4 v contour at Re = 1:

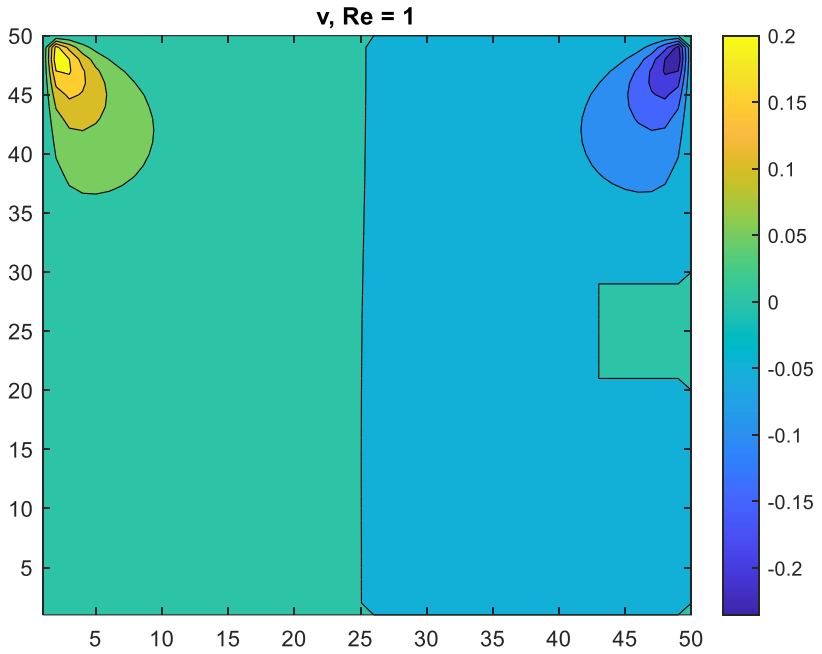


Figure 18: v contour at Re = 1

4-2-3 Contours of ψ, ω, u, and v at Re = 10:

4-2-3-1 ψ contour at Re = 10:

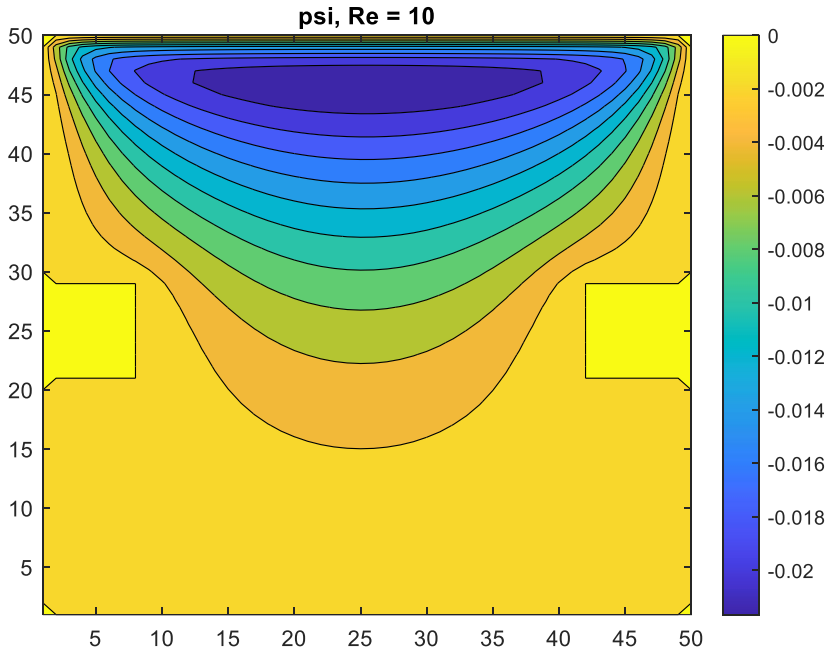


Figure 19: ψ contour at Re = 10

4-2-3-2 ω contour at Re = 10:

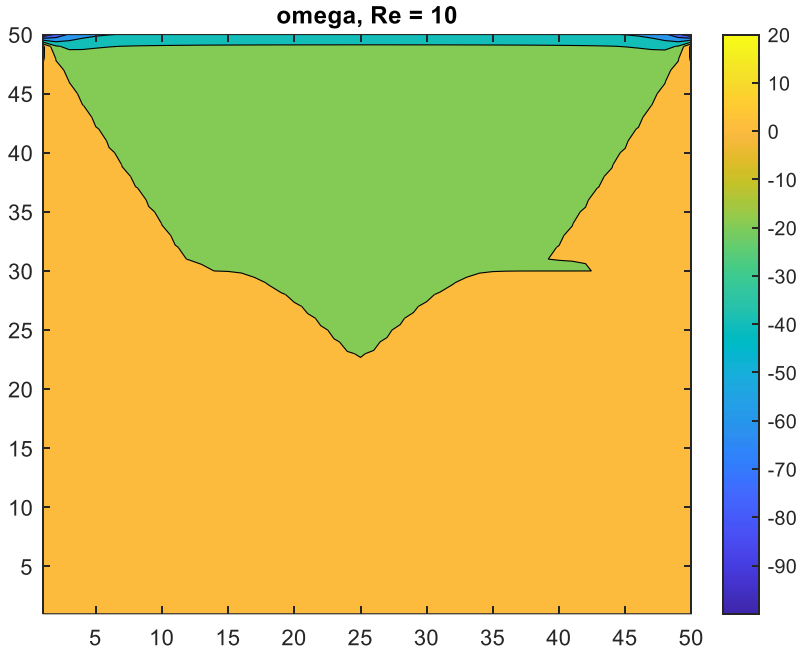


Figure 20: ω contour at Re = 10

4-2-3-3 u contour at Re = 10:

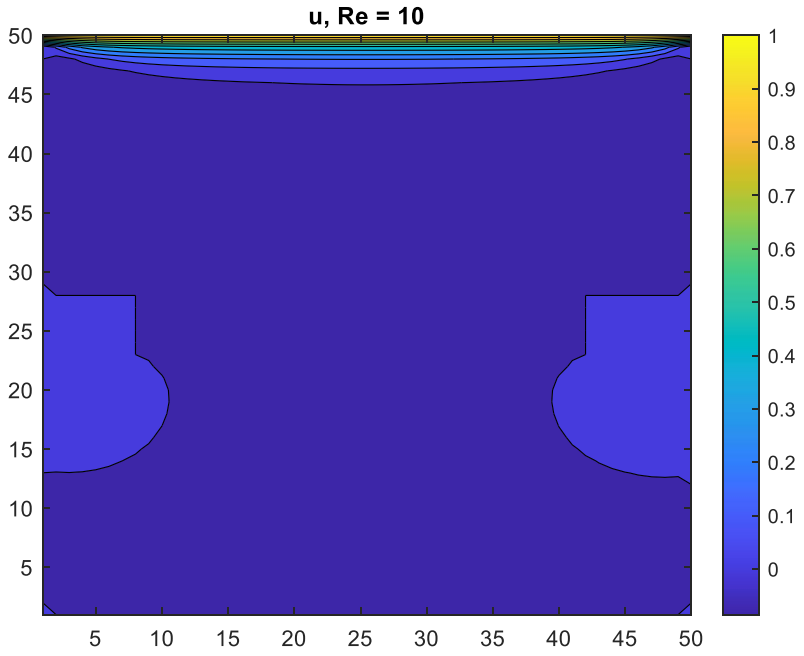


Figure 21: u contour at Re = 10

4-2-3-4 v contour at Re = 10:

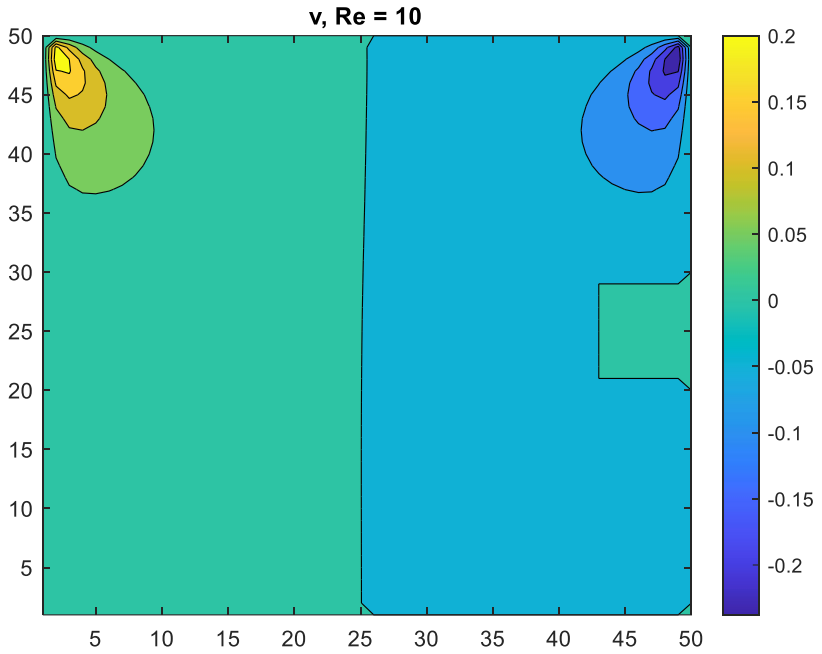


Figure 22: v contour at Re = 10

4-2-4 Contours of ψ, ω, u, and v at Re = 100:

4-2-4-1 ψ contour at Re = 100:

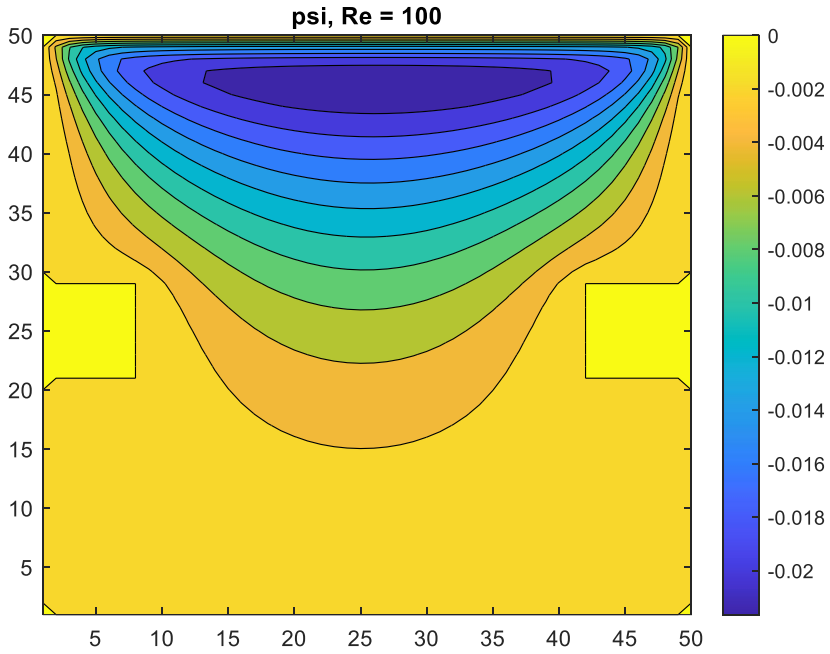


Figure 23: ψ contour at Re = 100

4-2-4-2 ω contour at Re = 100:

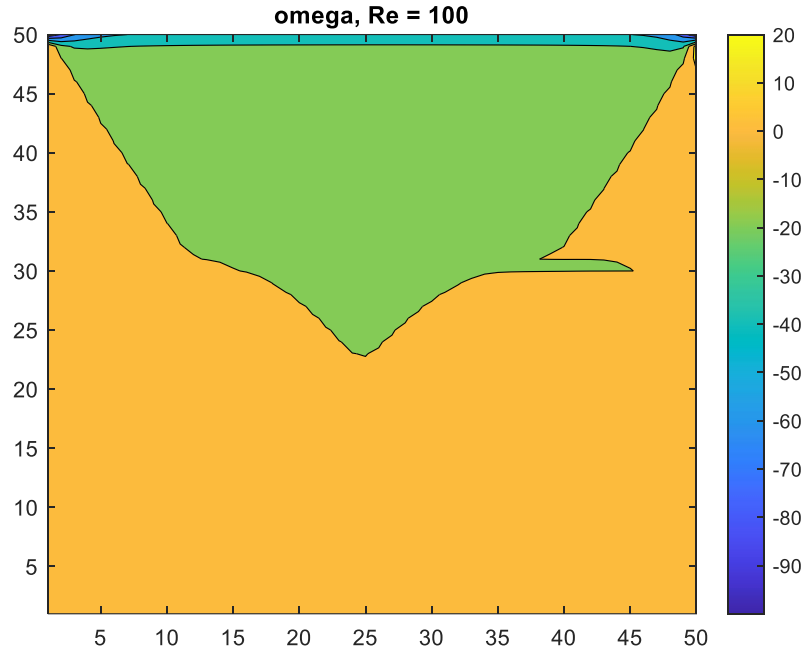


Figure 24: ω contour at Re = 100

4-2-4-3 u contour at Re = 100:

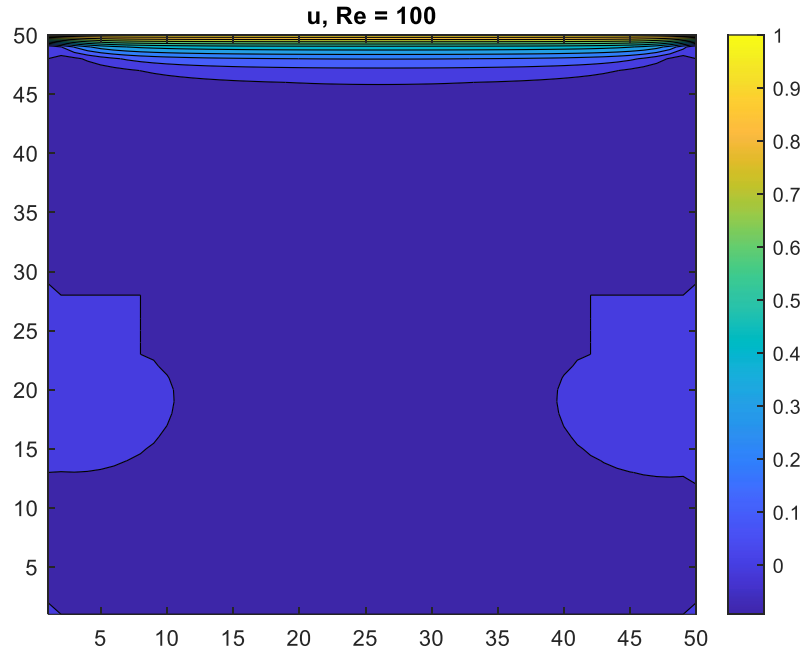


Figure 25: u contour at Re = 100

4-2-4-4 v contour at Re = 100:

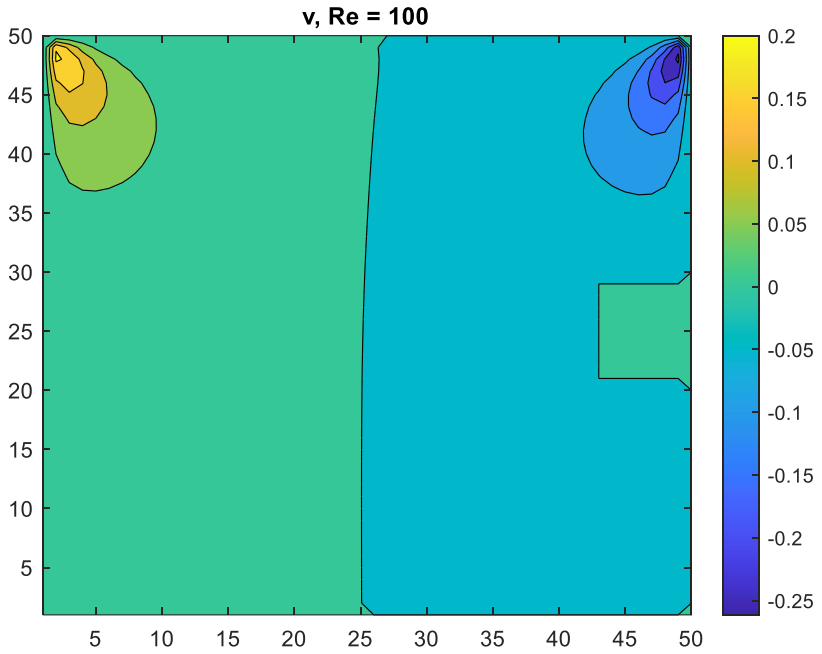


Figure 26: v contour at Re = 100

4-2-5 Task 3:

The solution time is measured using the MATLAB toc command and is displayed in the Command Window at the end of the computation:

Solution times for Re = 1, 10, and 100:

Re	Time (s)
1	9.471
10	9.8411
100	9.7126

4-2-6 Task 4:

Number of iterations required to reach convergence at Re = 1, 10, and 100:

Re	Number of Iterations
1	1126
10	1126
100	1128